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Software Design and Architecture Final Project

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1-Introduction

1.1 Purpose

This document specifies the software requirements and design for the Classroom Equipment Booking and Fault Tracking System (CEBFTS). It covers the complete functional and non-functional requirements, system design artifacts, and implementation details for all three iterations of the project.

This SRS is intended for the development team, course instructors, and evaluators. It documents all 15 use cases across three user roles (Teacher, Lab Manager, Technician) and describes how the system satisfies the project requirements including OOP principles, design patterns, database integration, and a three-layer architecture.

1.2 Product Scope

The Classroom Equipment Booking and Fault Tracking System is a Java-based desktop application that enables educational institutions to manage the lifecycle of classroom and lab equipment. The system allows teachers to search and book equipment, lab managers to approve/reject bookings and monitor inventory, and technicians to receive and resolve maintenance tasks.

The system eliminates manual paper-based booking processes, provides real-time equipment availability status, and ensures prompt resolution of faulty equipment through an integrated fault-reporting and maintenance workflow.

1.3 Title

Classroom Equipment Booking and Fault Tracking System (CEBFTS) — A three-layer Java application that automates equipment reservation, fault reporting, and maintenance management for academic labs and classrooms.

1.4 Objectives

- Provide a centralized platform for booking classroom and laboratory equipment.
- Enable teachers to search for available equipment and submit booking requests.
- Allow lab managers to approve/reject bookings and monitor overall equipment status.
- Facilitate fault reporting by teachers and assignment of maintenance tasks to technicians.
- Provide technicians with a structured workflow for tracking and resolving equipment faults.
- Maintain a complete audit log of all system actions for accountability and reporting.
- Apply OOP principles (encapsulation, inheritance, polymorphism, abstraction) and design patterns (Singleton, Facade, GRASP).

1.5 Problem Statement

In many educational institutions, the process of booking classroom equipment such as projectors, laptops, microscopes, and oscilloscopes is managed manually through paper forms or informal email requests. This approach leads to double-bookings, lack of real-time availability information, and delayed responses to booking requests. Furthermore, when equipment breaks down, there is no formal mechanism to report, track, and resolve faults efficiently.

This project addresses these problems by delivering a software system that automates equipment reservations and fault management. The system enforces role-based access, prevents scheduling conflicts, and creates a transparent pipeline from fault reporting to resolution.

The system can be easily built using Java 17, Maven, and SQL. It does not use any complex external frameworks and only uses basic Java database connectivity (JDBC). The system can run on any computer that has Java and SQL installed.

2-Overall Description

2.1- Product Perspective

CEBFTS is a standalone desktop application built from scratch as an academic project. It is not a component of a larger enterprise system. The application follows a three-layer architecture: a console-based UI layer (Main.java,CSS), a business logic/service layer (AuthService,

EquipmentService, FaultService, LabManagerService), and a data access layer (DAO classes backed by a SQL database).

The system interacts with a single external component — a SQL database server — via JDBC. All other functionality is self-contained within the Java application.

2.2- Product Functions

The following are the major functional groups of the system:

- Authentication: User login, registration, and password management.
- Equipment Management: Search equipment, check availability, update equipment status, monitor inventory.
- Booking Management: Create, approve, reject, and track booking requests.
- Fault Tracking: Report faulty equipment, assign maintenance tasks, update repair status, resolve faults.
- Audit Logging: Automatic logging of all significant system actions with user attribution.

2.3- List of Usecases

UC ID	Use Case Name	Role	Type
UC01-T	Search Equipments	Teacher	Unique
UC02-T	Check Availability	Teacher	Unique
UC03-T	Book Equipment	Teacher	Unique
UC04-T	Report Faulty Equipment	Teacher	Unique
UC05-T	Check Booking Status	Teacher	Read (CRUD)
UC01-LM	Approve/Reject Booking	Lab Manager	Unique
UC02-LM	Monitor Equipment Availability	Lab Manager	Unique
UC03-LM	Update Equipment Status	Lab Manager	Update (CRUD)
UC04-LM	View Reports and Analytics	Lab Manager	Unique

UC05-LM	Assign Maintenance Task	Lab Manager	Unique
UC01-TC	View Assigned Tasks	Technician	Read (CRUD)
UC02-TC	Update Repair Status	Technician	Unique
UC03-TC	Mark Fault as Resolved	Technician	Unique
UC04-TC	View Fault Details	Technician	Read (CRUD)
UC05-TC	View Maintenance History	Technician	Unique

2.4- Brief Usecases

1-Use case 1: Search Equipments

Use case Id: 1

Use case name: Search Equipments

Actors: Teacher

Type: Supporting/Secondary

Description: Teacher logs in the system and search his desired equipments. System runs the query in database and records the matching equipments. System shows the search results to teacher which contains the details of equipments.

2- Use case 2: Check availability of Equipments

Use case Id: 2

Use case name: Check Availability of Equipments

Actors: Teacher

Type: Supporting/Secondary

Description: Teacher searches his desired equipments and check the availability of equipments. System checks the status of equipment in database and veifies either it is available or booked. System shows the status of equipments in form of available or booked to teacher. If the equipment is available it shows the details of equipments as well.

3-Use case 3: Booking Equipments

Use case Id: 3

Use case name: Booking Equipments

Actors: Teacher

Type: Primary

Description: Teacher logins the system, search his desired equipments and submit booking request to the system. System records the booking request and update the equipment in reserved status. System shows the confirmation message to teacher which contains the details of equipments.

4-Use case 4: Report Faulty Equipments

Use case Id: 4

Use case name: Report Faulty Equipments

Actors: Teacher, Lab technician

Type: Primary

Description: Teacher selects the faulty equipment in the system and submit the faulty report. System saves the report in database and mark the equipment fault. System shows the complaint confirmation message to the teacher and notifies the report to further lab technician.

5- Use case 5: Check Booking Status

Use case Id: 5

Use case name: Check Booking Status

Actors: Teacher

Type: Supporting/Secondary

Description: Teacher logs in the system and checks the status of his previously submitted booking request. System verifies the current status of booking in database. System shows the teacher either the booking is approved, rejected or pending. If the system approves the request it shows the details of equipment to the teacher as well.

6- Approve/Reject Booking

Actors: Lab Manager

Type: Primary

The Lab Manager reviews equipment booking requests submitted by teachers and evaluates them based on equipment availability, scheduling constraints, and system rules. The manager then makes a decision to either approve or reject the booking request to ensure fair and efficient utilization of classroom resources and to avoid scheduling conflicts.

7-Monitor Equipment Availability

Actors: Lab Manager

Type: Primary

The Lab Manager monitors the real-time availability status of all classroom equipment within the system. This use case helps ensure proper allocation of resources, prevents double booking, and provides an updated view of equipment status to support effective decision-making.

8- Update Equipment Status

Actors: Lab Manager

Type: Primary

The Lab Manager updates the status of classroom equipment based on its current condition or usage. The status may reflect whether the equipment is available, booked, faulty, under repair, or retired. This ensures accurate tracking of equipment lifecycle and supports maintenance planning.

9- View Reports and Analytics

Actors: Lab Manager

Type: Primary

The Lab Manager accesses system-generated reports and analytics related to equipment usage, fault frequency, booking trends, and maintenance activities. These insights assist in evaluating system performance and support informed decision-making for resource management.

10- Assign Maintenance Tasks

Actors: Lab Manager
Type: Primary

The Lab Manager assigns maintenance or repair tasks to technicians based on reported equipment faults, severity level, and priority. This ensures that issues are addressed in a timely manner and that faulty equipment is restored efficiently for classroom use.

11- View Assigned Maintenance Tasks

Actors: Technician
Type: Primary

Description:

The Technician logs into the system and opens the maintenance dashboard. The system displays a list of all tasks assigned to the Technician along with details such as equipment name, fault description, severity level, and current status. The Technician reviews the tasks to decide which ones to handle first.

12-Update Repair Status

Actors: Technician
Type: Primary

Description:

The Technician selects a specific maintenance task from the list and updates its repair status to Pending, In Progress, or Completed. The system saves the updated status and reflects the change in the task list so that progress can be tracked by other users.

13-Mark Fault as Resolved

Actors: Technician
Type: Primary

Description:

After completing the repair, the Technician opens the task and marks the fault as resolved. The system updates the fault record, changes the equipment status to available, and records the completion in the system.

14- View Fault Details

Actors: Technician
Type: Secondary

Description:

The Technician selects a fault report to view its details. The system displays complete information including fault description, severity level, reporting date, and related equipment, helping the Technician understand the issue before starting the repair.

15-View Maintenance History

Actors: Technician

Type: Secondary

Description:

The Technician selects an equipment item to check its maintenance history. The system shows previous repair records, including past faults, dates, and actions taken, allowing the Technician to review earlier issues related to the equipment.

2.5- Extended Usecases

1-Use case 1: Search Equipments

- **Use case Id: 1**
- **Use Case Name:** Search Equipments
- **Scope:** Classroom Equipment Booking and fault tracking Management System
- **Level:** Sub Function
- **Primary Actor:** Teacher
- **Stakeholders and Interests:**
 - 1-Teacher:** Teacher wants to easily search and book required equipment and quickly report faulty equipment to ensure smooth teaching activities.
 - 2-Lab Manager:** Lab Manager wants to efficiently manage equipment requests, approve or reject bookings, and ensure faulty equipment is assigned to technicians for timely repair.
 - 3-Technician :** Technician wants to receive fault reports and update repair status to ensure equipment is fixed and available for use.
- **Preconditions:**
 - 1-Teacher** should be registered and logged in to the system.
 - 2-System** should be operational.
 - 3-Teacher** should have access to search equipments.
 - 4-Equipment** data should be available in the database.
- **Success Guarantee (Post Conditions):**
 - Search results are displayed to the teacher
 - Relevant equipment information is retrieved from the database**
 - System remains unchanged
- **Main Success Scenario (two-column format)**

Actor Action	System Response
Teacher logs into the system	System authenticates the teacher and grants access
Teacher enters equipment name/type in search bar	System receives the search query
Teacher submits the search request	System processes the request

	System searches the database for matching equipment
	System retrieves matching equipment records
	System displays search results with equipment details to the teacher

- **Extensions**

If the teacher enters invalid or empty input, then the system will display an error message.

If no matching equipment is found, then the system will display “No results found”.

If the system fails to access the database, then the system will show an error message.

If the session expires, then the system will redirect the teacher to the login page.

- **Special Requirements**

1. System must respond to a search request within 2 seconds.
2. Teacher must have valid login credentials to perform a search.
3. Only authorized teachers can search certain equipment categories

- **Technology and Data Variations List**

1- System should support desktop and mobile browsers.

2- Equipment data may include images, text, or PDF manuals.

3- Database may contain hundreds of equipment records.

- **Frequency of Occurrence:** High, It is of daily, multiple times use per teacher.

- **Miscellaneous**

1-System should log all search queries for audit purposes.

2- Equipment booking confirmation can be emailed to the teacher.

4- System must maintain previous 6 months of booking and fault records.

2- Use case 2: Check availability of Equipments

- **Use case Id: 2**

- **Use Case Name:** Check availability of Equipments

- **Scope:** Classroom Equipment Booking and fault tracking Management System

- **Level:** Sub Function

- **Primary Actor:** Teacher

- **Stakeholders and Interests:**

1-Teacher: Teacher wants to easily search and book required equipment also wants to check the availability and quickly report faulty equipment to ensure smooth teaching activities.

2-Lab Manager: Lab Manager wants to efficiently manage equipment requests, approve or reject bookings, and ensure faulty equipment is assigned to technicians for timely repair.

3-Technician : Technician wants to receive fault reports and update repair status to ensure equipment is fixed and available for use.

- **Preconditions**

- Teacher is logged into the system.

- Teacher is registered in the system.

- Equipment data is available in the database.

- Teacher has permission to check equipment status.

- **Success Guarantee (Post Conditions)**

- 1-Equipment availability status is displayed to the teacher.

- 2- System fetches current status from the database.

- 3- No changes are made to system data.

- **Main Success Scenario (two-column format)**

Actor Action	System Response
Teacher searches for desired equipment	System receives the request
Teacher selects equipment to check availability	System checks the database for current status
	System verifies if equipment is available or booked
	System displays availability status and equipment details to teacher

- **Extensions**

- 1- If the teacher enters invalid or empty input, then the system displays an error message.

- 2- If the equipment does not exist in the database, then the system displays “Equipment not found.”

- 3- If the system fails to access the database, then an error message is displayed.

- 4- If the session expires, then the system redirects the teacher to the login page.

- **Special Requirements**

- 1- System must respond within 2 seconds.

- 2-Teacher must have valid login credentials to check availability.

- 3- Only authorized teachers can check availability for certain equipment categories.

- **Technology and Data Variations List**

- System should support desktop and mobile browsers.

- Database may contain large volume of equipment records.

- Equipment information may include text, images, or PDF manuals

- **Frequency of Occurrence:** High
Teachers may check equipment availability multiple times daily before booking
- **Miscellaneous**
 - 1- All availability checks should be logged for auditing purposes.
 - 2- System must display the most recent equipment status to avoid double booking.

3-Use case 3: Booking Equipments

- **Use case Id:** 3
- **Use Case Name:** Booking Equipments
- **Scope:** Classroom Equipment Booking and fault tracking Management System
- **Level:** User-goal
- **Primary Actor:** Teacher
- **Stakeholders and Interests:**
 - 1-Teacher:** Teacher wants to easily search and book required equipment and quickly report faulty equipment to ensure smooth teaching activities.
 - 2-Lab Manager:** Lab Manager wants to efficiently manage equipment requests, approve or reject bookings, and ensure faulty equipment is assigned to technicians for timely repair.
 - 3-Technician :** Technician wants to receive fault reports and update repair status to ensure equipment is fixed and available for use.
- **Preconditions**
 - 1- Teacher is logged into the system.
 - 2- Teacher is registered in the system.
 - 3- Equipment data is available in the database.
 - 4- Teacher has permission to book equipment
- **Success Guarantee (Post Conditions)**

Equipment booking is recorded in the system.

Equipment status is updated to “Reserved”.

Teacher receives booking confirmation with equipment details.

Lab Manager is notified of new booking for approval

- **Main Success Scenario (two-column format)**

Actor Action	System Response
Teacher searches for desired equipment	System displays list of available equipment
Teacher selects equipment and submits booking request	System records the booking request in the database

	System updates equipment status to “Reserved”
	System shows confirmation message with equipment details to the teacher
	System notifies Lab Manager for approval

- **Extensions**

- 1- If the selected equipment is already booked, then the system displays “Equipment already booked”.
- 2- If the teacher enters invalid input, then the system displays an error message.
- 3- If the system fails to update the database, then the system shows an error message.
- 4- If the session expires, then the system redirects the teacher to the login page.

- **Special Requirements**

- 1- System must respond within 2 seconds.
- 2- Teacher must have valid login credentials to book equipment.
- 3- Only authorized teachers can book equipment

- **Technology and Data Variations List**

- 1- System should support desktop and mobile browsers.
- 2- Equipment data may include text, images, or PDF manuals.
- 3- Database may contain hundreds of equipment records

- **Frequency of Occurrence**

Low

Teachers may book equipment 2–3 times per week for classroom use.

- **Miscellaneous**

- 1- Booking confirmation should be logged for auditing purposes.
- 2- System should prevent double booking of the same equipment at the same time.
- 3- Lab Manager should be able to approve or reject bookings based on equipment availability.

4-Use case 4: Report Faulty Equipments

- **Use case Id: 4**

- **Use Case Name:** Report Faulty Equipments

- **Scope:** Classroom Equipment Booking and fault tracking Management System

- **Level:** User-goal
- **Primary Actor:** Teacher
- **Stakeholders and Interests:**
 - 1-Teacher:** Teacher wants to easily search and book required equipment and quickly report faulty equipment to ensure smooth teaching activities.
 - 2-Lab Manager:** Lab Manager wants to efficiently manage equipment requests, approve or reject bookings, and ensure faulty equipment is assigned to technicians for timely repair.
 - 3-Technician :** Technician wants to receive fault reports and update repair status to ensure equipment is fixed and available for use.
- **Preconditions**
 - 1- Teacher is logged into the system.
 - 2- Teacher is registered in the system.
 - 3- Equipment data is available in the database.
 - 4- Teacher has permission to report faulty equipment
- **Success Guarantee (Post Conditions)**
 - 1-Fault report is recorded in the system database.
 - 2-Equipment is marked as faulty in the system.
 - 3-Teacher receives a confirmation message for the reported fault.
 - 4-Lab Technician is notified about the faulty equipment
- **Main Success Scenario (two-column format)**

Actor Action	System Response
Teacher searches and selects the faulty equipment	System displays equipment details
Teacher submits a fault report	System records the fault report in the database
	System marks the equipment status as “Faulty”
	System shows a confirmation message to the teacher
	System notifies Lab Technician for further action

- **Extensions**

- 1- If the teacher submits a report without selecting equipment, then the system displays “Please select equipment to report”.
- 2- If the system fails to save the fault report, then an error message is displayed.
- 3- If the teacher enters invalid details (e.g., description too short), then the system prompts for valid input.
- 4- If the session expires, then the teacher is redirected to the login page

- **Special Requirements**

- 1-System must respond within 2 seconds.
- 2-Teacher must have valid login credentials to report faulty equipment.
- 3-Only authorized teachers can submit fault reports

- **Technology and Data Variations List**

- 1-System should support desktop and mobile browsers.
- 2-Fault report may include text description, images, or documents (PDF).
- 3- Database may contain hundreds of equipment records

- **Frequency of Occurrence**

- 1-Teachers may report faulty equipment 1–2 times per month.
- 2- Lab Technician may receive multiple reports per week

- **Miscellaneous**

- 1-All fault reports should be logged for auditing purposes.
- 2-System should allow Lab Technician to update repair status once completed.
- 3- Faulty equipment should be blocked from booking until repaired.

Use case 5: Check Booking Status

- **Use case Id: 5**
- **Use Case Name:** Check Booking Status
- **Scope:** Classroom Equipment Booking and fault tracking Management System
- **Level:** Sub Function

- **Primary Actor:** Teacher
- **Stakeholders and Interests:**
 - 1-Teacher :** Teacher wants to easily search and book required equipment also wants to check the booking status either it is approved, rejected or pending and quickly report faulty equipment to ensure smooth teaching activities.
 - 2-Lab Manager:** Lab Manager wants to efficiently manage equipment requests, approve or reject bookings, and ensure faulty equipment is assigned to technicians for timely repair.
 - 3-Technician :** Technician wants to receive fault reports and update repair status to ensure equipment is fixed and available for use.
- **Preconditions**

- 1-Teacher is logged into the system.
- 2-Teacher is registered in the system.
- 3-Booking request data exists in the database
- 4-Teacher has permission to view booking status

- **Success Guarantee (Post Conditions)**
 - 1- Booking status is displayed to the teacher.
 - 2- System retrieves current booking status from the database.
 - 3- No changes are made to system data.
 - 4- Teacher can view booking details if the request is approved
- **Main Success Scenario (two-column format)**

Actor Action	System Response
Teacher logs into the system	System authenticates the teacher
Teacher selects option to check booking status	System receives the request
Teacher views previously submitted booking requests	System retrieves booking records from the database
	System checks current status of each booking
	System displays booking status (Approved / Rejected / Pending) to the teacher
	System shows equipment details if booking is approved

- **Extensions**

- 1-If no booking request exists, then the system displays “No booking found”.
- 2-If the system fails to retrieve data from database, then an error message is displayed.
- 4- If the session expires, then the system redirects the teacher to the login page.
- 5-If invalid request is made, then the system shows an error message

- **Special Requirements**

- 1-System must respond within 2 seconds
- 2-Teacher must have valid login credentials to check booking status.
- 3-Only authorized teachers can view their booking status

- **Technology and Data Variations List**

- 1- System should support desktop and mobile browsers
- 2-Booking data may include equipment details, date, and time.
- 3- Database may contain multiple booking records

- **Frequency of Occurrence**

Teachers may check booking status multiple times after submitting a request

- **Miscellaneous**

- 1-System should display the most updated booking status
- 2-All booking status checks should be logged for auditing
- 3-System should maintain history of previous booking records.

6- Approves or Rejects Booking

Use Case ID: UC01

Use Case Name: Approve or Reject Booking

Scope: Classroom Equipment Booking and Fault Tracking System

Level: User Goal**Primary Actor: Lab Manager****Stakeholders and Interests:**

- The Lab Manager wants the system to ensure conflict-free scheduling and efficient resource allocation.
- The Teacher wants the system to provide quick approval so that they can plan their classes effectively.
- The Students want the system to ensure uninterrupted classes with the required equipment available.
- The Administration wants the system to optimize resource utilization and reduce wastage.
- The IT Department wants the system to maintain reliability, security, and proper activity logging.

Preconditions:

- The Teacher has submitted a booking request in the system.
- The Lab Manager has successfully authenticated into the system.
- The system has stored the equipment information in the database.

Success Guarantee (Post Conditions):

- The Lab Manager has approved or rejected the booking request.
- The system has updated the booking status in the schedule.
- The system has sent a notification to the Teacher.
- The system has stored the decision in the activity logs.

Main Success Scenario

User Action	System Action
The Lab Manager logs into the system.	The system authenticates the Lab Manager credentials.
The Lab Manager opens the booking requests page.	The system retrieves and displays all pending booking requests.

The Lab Manager filters or searches the booking requests.	The system displays the filtered list of booking requests.
The Lab Manager selects a specific booking request.	The system displays detailed booking information.
The Lab Manager checks equipment availability.	The system performs conflict detection on the selected time slot.
The Lab Manager approves or rejects the booking request.	The system updates the booking status accordingly.
	The system updates the equipment schedule.
	The system sends a notification to the Teacher.
	The system records the action in the system logs.

Extensions:

- If the system detects a time slot conflict, the system blocks the approval action.
- If the equipment is marked as faulty, the system prevents the booking from being approved.
- If the system detects a duplicate booking request, the system displays a warning message.
- If a network failure occurs during the process, the system does not save the action.
- If an unauthorized user attempts access, the system denies the operation.

Special Requirements:

- The system must respond within 2 seconds for booking operations.
- The system must perform real-time conflict detection.
- The system should be responsive 24/7.

- The system must maintain secure authentication and logging mechanisms.

Technology and Data Variations List:

- The system uses a web-based user interface for interaction.
- The system stores booking data in a relational database.
- The system sends notifications through email or in-system alerts.

Frequency of Occurrence:

- The Lab Manager performs this use case multiple times daily.

Open Issues:

- The system may need to support automatic approval for certain bookings.
- The system may need to introduce priority-based booking features.
- The system may need to handle emergency booking scenarios.
- The system may need integration with the institutional timetable system.
- The system may need to decide between email and SMS notifications.

7- Monitors Equipment Availability

Use Case ID: UC02

Use Case Name: Monitor Equipment Availability

Scope: Classroom Equipment Booking System

Level: User Goal

Primary Actor: Lab Manager

Stakeholders and Interests:

- The Lab Manager wants the system to provide real-time visibility of equipment status.
- The Teacher wants the system to show accurate availability information.
- The Students want the system to ensure that equipment is ready for use.
- The Administration wants the system to track equipment utilization.
- The IT Department wants the system to perform efficiently.

Preconditions:

- The system has registered equipment in the database.
- The Lab Manager has logged into the system.

Success Guarantee:

- The system displays the current availability status of all equipment.

Main Success Scenario

User Action	System Action
The Lab Manager logs into the system.	The system authenticates the Lab Manager.
The Lab Manager opens the dashboard.	The system retrieves equipment data.
The Lab Manager applies filters such as type or date.	The system filters the equipment list accordingly.
The Lab Manager views equipment availability.	The system displays status such as Available, Booked, or Faulty.
The Lab Manager refreshes the dashboard.	The system updates and displays real-time data.

Extensions:

- If the system fails to load data, the system displays an error message.
- If the server response is delayed, the system shows a loading indicator.
- If the user applies an invalid filter, the system prompts for correction.
- If equipment data is missing, the system flags an inconsistency.
- If unauthorized access occurs, the system denies access.

Special Requirements:

- The system must provide real-time updates.
- The system must ensure fast dashboard rendering.
- The system must provide a user-friendly interface.
- The system should be responsive 24/7.

Technology and Data Variations List:

- The system categorizes equipment status into predefined types.
- The system supports filtering and searching mechanisms.

Frequency of Occurrence:

- The Lab Manager performs this action frequently during daily operations.

Open Issues:

- The system may need to support mobile-friendly dashboards.
- The system may need advanced filtering options.
- The system may need an auto-refresh feature.
- The system may need integration with booking modules.
- The system may need graphical dashboards.

8- Updates Equipment Status

Use Case ID: UC03

Use Case Name: Update Equipment Status

Scope: Classroom Equipment Booking and Fault Tracking System

Level: User Goal

Primary Actor: Lab Manager

Stakeholders and Interests:

- The Lab Manager wants the system to maintain accurate and up-to-date equipment status.
- The Technician wants the system to reflect the correct repair status of equipment.

- The Teacher wants the system to ensure that only functional equipment is available for booking.
- The Administration wants the system to track equipment lifecycle and condition.
- The IT Department wants the system to maintain data consistency and integrity.

Preconditions:

- The system has registered the equipment in the database.
- The Lab Manager has successfully authenticated into the system.
- The system has stored the current status of the equipment.

Success Guarantee:

- The Lab Manager has updated the equipment status successfully.
- The system has stored the updated status in the database.
- The system has reflected the updated status across all modules.
- The system has recorded the update action in the logs.

Main Success Scenario

User Action	System Action
The Lab Manager logs into the system.	The system authenticates the Lab Manager.
The Lab Manager selects an equipment item.	The system retrieves equipment details.
The Lab Manager reviews the current status.	The system displays the current status.
The Lab Manager selects a new status.	The system validates the selected status.
The Lab Manager confirms the update.	The system saves the updated status.

	The system updates the status across all modules.
	The system logs the update action.

Extensions:

- If the system detects an invalid status, the system displays an error.
- If the system cannot find the equipment, the system shows an alert.
- If concurrent updates occur, the system shows a conflict warning.
- If the system fails, the system rolls back the changes.
- If an unauthorized user attempts access, the system denies the action.

Special Requirements:

- The system must enforce strict data validation rules.
- The system must maintain audit logs.
- The system must ensure data consistency across modules.
- The system should be responsive every time.

Open Issues:

- The system may need to allow multiple roles to update status.
- The system may need approval workflows for status changes.
- The system may need automatic updates from technicians.
- The system may need to maintain status history.
- The system may need to notify users about status changes.

9- Views Reports and Analytics

Use Case ID: UC04

Use Case Name: View Reports and Analytics

Scope: Classroom Equipment Booking and Fault Tracking System

Level: User Goal

Primary Actor: Lab Manager

Stakeholders and Interests:

- The Lab Manager wants the system to provide meaningful insights for decision-making.
- The Administration wants the system to generate reports for performance evaluation and planning.
- The IT Department wants the system to ensure report accuracy and data integrity.
- The Finance Department wants the system to analyze cost efficiency and maintenance expenses.
- The Auditors want the system to maintain historical records for auditing purposes.

Preconditions:

- The system has stored sufficient operational data.
- The Lab Manager has successfully logged into the system.
- The reporting module is available and functional.

Success Guarantee :

- The system has generated the requested report successfully.
- The system has displayed the report in a readable format.
- The system has optionally exported the report if requested.
- The system has logged the report generation activity.

Main Success Scenario

User Action	System Action
The Lab Manager logs into the system.	The system authenticates the Lab Manager credentials.
The Lab Manager navigates to the reports section.	The system loads available report options.
The Lab Manager selects a report type.	The system retrieves relevant data from the database.
The Lab Manager applies filters such as date or category.	The system processes the selected filters.
The Lab Manager requests report generation.	The system generates the report based on selected criteria.

The Lab Manager views the generated report.	The system displays the report using charts and tables.
The Lab Manager chooses to export the report.	The system generates a downloadable file (PDF/Excel).
	The system logs the report generation activity.

Extensions:

- If the system finds no data for the selected filters, the system displays an empty report.
- If the dataset is too large, the system displays a delay message.
- If report generation fails, the system prompts the user to retry.
- If the user enters invalid filters, the system asks for correction.
- If a system error occurs, the system displays an error message.

Special Requirements:

- The system must support graphical visualization such as charts and graphs.
- The system must provide export options such as PDF and Excel formats.
- The system must generate reports efficiently within acceptable response time.
- The system should be responsive 24/7.

Technology and Data Variations List:

- The system uses a database to retrieve historical and real-time data.
- The system supports multiple report types such as usage reports and fault reports.
- The system provides export functionality in different file formats.

Frequency of Occurrence:

- The Lab Manager performs this use case weekly or monthly.

Open Issues:

- The system may need to support real-time analytics.
- The system may need to allow custom report creation.
- The system may need additional export formats.
- The system may need a defined data retention policy.
- The system may need integration with external reporting tools.

10- Assigns Maintenance Tasks

Use Case ID: UC05

Use Case Name: Assign Maintenance Tasks

Scope: Classroom Equipment Booking and Fault Tracking System

Level: User Goal

Primary Actor: Lab Manager

Stakeholders and Interests:

- The Lab Manager wants the system to ensure timely and efficient fault resolution.
- The Technician wants the system to provide clear and prioritized task assignments.
- The Teacher wants the system to ensure quick repair of faulty equipment.
- The Students want the system to ensure availability of functional equipment.
- The Administration wants the system to minimize equipment downtime.

Preconditions:

- The system has recorded a fault in the system.
- The Lab Manager has successfully logged into the system.
- The system has registered technicians in the database.

Success Guarantee:

- The system has assigned the maintenance task to a technician.
- The system has set the task priority based on severity.
- The system has notified the assigned technician.
- The system has recorded the assignment in logs.

Main Success Scenario

User Action	System Action
The Lab Manager logs into the system.	The system authenticates the Lab Manager credentials.
The Lab Manager navigates to the fault management section.	The system retrieves and displays all reported faults.

The Lab Manager sorts faults based on severity.	The system organizes faults according to priority levels.
The Lab Manager selects a fault.	The system displays detailed fault information.
The Lab Manager selects a technician.	The system verifies technician availability.
The Lab Manager assigns the maintenance task.	The system creates a maintenance task entry.
	The system assigns priority based on fault severity.
	The system notifies the technician.
	The system logs the task assignment.

Extensions:

- If no technician is available, the system displays a warning message.
- If the fault has high severity, the system automatically assigns high priority.
- If a duplicate assignment is attempted, the system prevents duplication.
- If the system fails during assignment, the system does not save the task.
- If the selected technician is overloaded, the system suggests another technician.

Special Requirements:

- The system must support priority-based task scheduling.
- The system should be responsive 24/7.
- The system must provide real-time notification to technicians.
- The system must track maintenance tasks effectively.

Technology and Data Variations List:

- The system stores fault and task data in a centralized database.
- The system categorizes faults into severity levels such as Low, Medium, and High.
- The system uses notification services such as email or system alerts.

Frequency of Occurrence:

- The Lab Manager performs this use case regularly depending on fault occurrences.

Open Issues:

- The system may need to support automatic task assignment.
- The system may need workload balancing among technicians.
- The system may need escalation rules for unresolved tasks.
- The system may need defined service-level agreements (SLAs).
- The system may need integration with maintenance history logs.

11- View Assigned Maintenance Tasks

Use Case ID: UC01

Use Case Name: View Assigned Maintenance Tasks

Scope: Classroom Equipment Booking and Fault Tracking System

Level: User Goal

Primary Actor: Technician

Stakeholders and Interests:

- The Technician wants to view all assigned maintenance tasks clearly to manage workload.
- The Lab Manager wants technicians to complete tasks on time.
- The Teacher wants faulty equipment to be repaired quickly.
- The Administration wants efficient maintenance and reduced downtime.
- The IT Department wants smooth system performance and accurate data retrieval.

Preconditions:

- The Technician has a valid account in the system.
- The Technician has successfully logged into the system.
- Faults have been reported and assigned to the Technician.

Success Guarantee (Post Conditions):

- The Technician can view all assigned maintenance tasks.
- The system displays task details including equipment name, severity, and status.
- No data is altered during this process.

Main Success Scenario

User Action	System Action
-------------	---------------

The Technician logs into the system.	The system authenticates the Technician credentials.
The Technician navigates to the maintenance dashboard.	The system retrieves assigned tasks from the database.
The Technician selects "View Tasks".	The system displays a list of assigned tasks.
The Technician views task details.	The system shows equipment name, fault severity, and status.

Extensions:

- If login credentials are incorrect, the system denies access.
- If no tasks are assigned, the system displays a "No Tasks Available" message.
- If a network issue occurs, the system fails to load tasks.
- If unauthorized access is attempted, the system blocks access.

Special Requirements:

- The system must respond within 2 seconds.
- The interface should be user-friendly and easy to navigate.
- The system must ensure secure authentication.

Technology and Data Variations List:

- Web-based dashboard interface.
- Data stored in a centralized database.

Frequency of Occurrence:

Performed multiple times daily.

Open Issues:

- Task prioritization visualization not defined.
- Filtering and sorting options may be needed.

12- Update Repair Status

Use Case ID: UC02

Use Case Name: Update Repair Status

Scope: Classroom Equipment Booking and Fault Tracking System

Level: User Goal

Primary Actor: Technician

Stakeholders and Interests:

- The Technician wants to update repair progress easily.
- The Lab Manager wants real-time updates on maintenance.
- The Administration wants accurate maintenance records.
- The IT Department wants data consistency and integrity.

Preconditions:

- The Technician is logged into the system.
- A maintenance task is assigned to the Technician.

Success Guarantee (Post Conditions):

- The repair status is successfully updated.
- The system stores updated status in the database.
- The system reflects changes in real-time.

Main Success Scenario

User Action	System Action
The Technician opens the task list.	The system displays assigned tasks.
The Technician selects a task.	The system shows task details.
The Technician selects a new status (Pending/In Progress/Completed).	The system validates the status.
The Technician confirms the update.	The system updates the status in the database.
	The system reflects the updated status on dashboard.

Extensions:

- If invalid status is selected, the system shows an error.
- If database error occurs, the system does not save changes.
- If network failure occurs, update is not completed.

Special Requirements:

- Status update must be real-time.
- Only valid status transitions should be allowed.

Technology and Data Variations List:

- Dropdown menu for status selection.
- Database update through backend system.

Frequency of Occurrence:

Frequently during maintenance tasks.

Open Issues:

- Automatic notifications after update not defined.
- Audit logs for changes may be required.

13- Mark Fault as Resolved

Use Case ID: UC03

Use Case Name: Mark Fault as Resolved

Scope: Classroom Equipment Booking and Fault Tracking System

Level: User Goal

Primary Actor: Technician

Stakeholders and Interests:

- The Technician wants to close completed tasks.
- The Lab Manager wants equipment to be available again.
- The Teacher wants equipment ready for use.
- The Administration wants reduced downtime.

Preconditions:

- The Technician has completed the repair.
- The Technician is logged into the system.

Success Guarantee (Post Conditions):

- Fault is marked as resolved.
- Equipment status is updated to available.
- The system records the completion.

Main Success Scenario

User Action	System Action
The Technician selects a completed task.	The system displays task details.
The Technician clicks "Mark as Resolved".	The system verifies task completion.
The Technician confirms the action.	The system updates fault status.
	The system updates equipment availability.
	The system logs the action.

Extensions:

- If task is incomplete, system blocks resolution.
- If system error occurs, action is not saved.
- If unauthorized access, system denies action.

Special Requirements:

- Confirmation prompt must be shown.
- Data must be saved securely.

Technology and Data Variations List:

- Button-based UI action.
- Database update for fault and equipment.

Frequency of Occurrence:

Occasional (after repairs).

Open Issues:

- Verification mechanism not defined.
- Notification to teacher not specified.

14-View Fault Details

Use Case ID: UC04

Use Case Name: View Fault Details

Scope: Classroom Equipment Booking and Fault Tracking System

Level: Sub-function

Primary Actor: Technician

Stakeholders and Interests:

- The Technician wants complete fault information before repair.
- The Lab Manager wants accurate diagnosis.

Preconditions:

- Fault has been reported.
- Technician is logged in.

Success Guarantee (Post Conditions):

Fault details are displayed correctly.

Main Success Scenario

User Action	System Action
The Technician selects a fault report.	The system retrieves fault data.
The Technician opens fault details.	The system displays severity, description, and date.

Extensions:

- If fault not found, system shows error.
- If system fails, data not loaded.

Special Requirements:

Information must be clearly displayed.

Technology and Data Variations List:

Detail view interface.

Frequency of Occurrence:

Frequently before repair.

Open Issues:

Image/file attachment support is not defined.

15-View Maintenance History

Use Case ID: UC05

Use Case Name: View Maintenance History

Scope: Classroom Equipment Booking and Fault Tracking System

Level: Sub-function

Primary Actor: Technician

Stakeholders and Interests:

- The Technician wants to review past repairs.
- The Administration wants historical records.

Preconditions:

- Equipment exists in system.

- Technician is logged in.

Success Guarantee (Post Conditions):

Maintenance history is displayed.

Main Success Scenario

User Action	System Action
The Technician selects equipment.	The system retrieves maintenance history.
The Technician views records.	The system displays past repairs and logs.

Extensions:

- If no history exists, system shows empty message.
- If system fails, data not displayed.

Special Requirements:

Data should load quickly.

Technology and Data Variations List:Table format display.

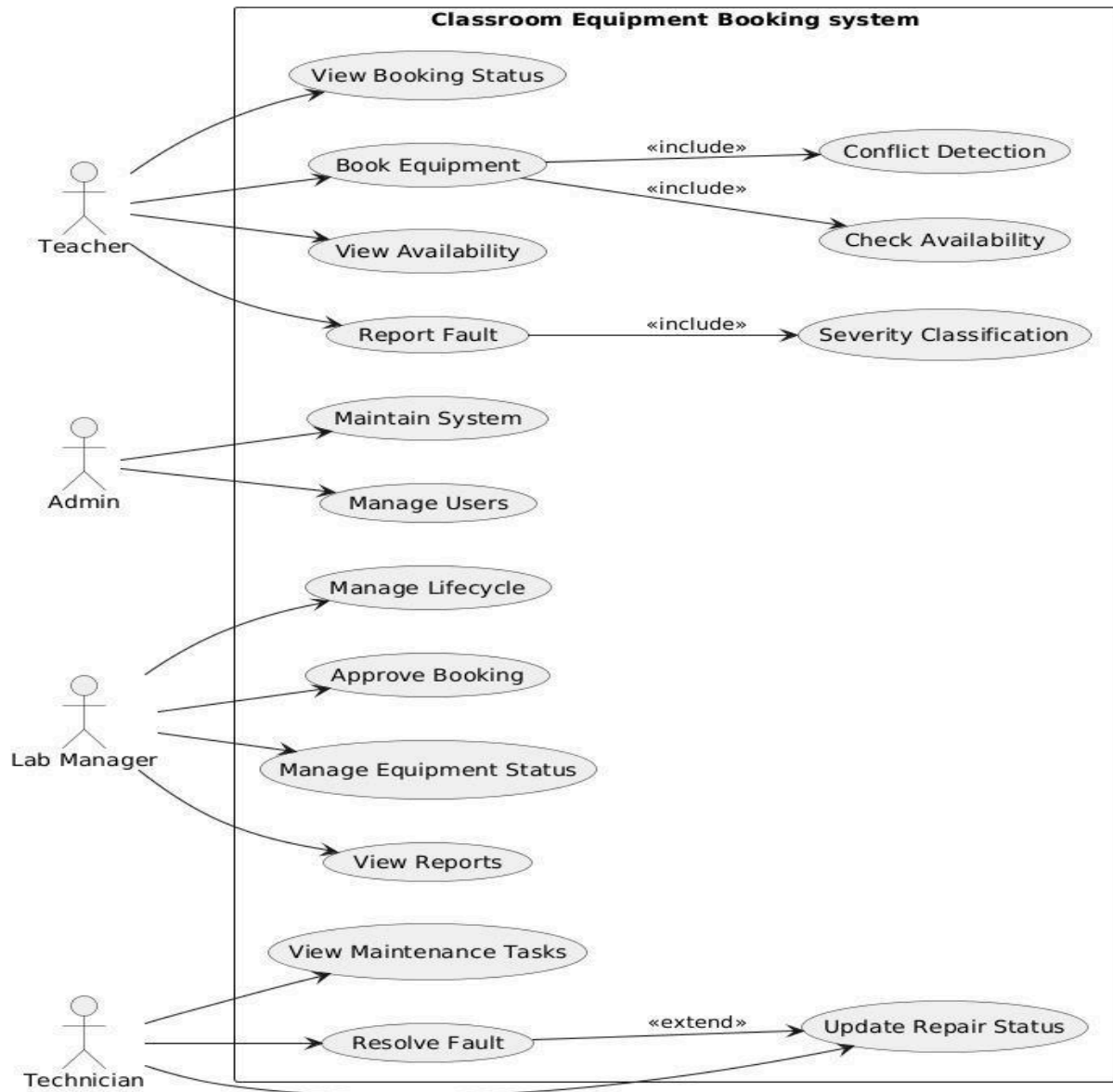
Frequency of Occurrence:

Occasionally.

Open Issues:

- Filtering options not defined.
- Sorting of records not specified.

2.5- Usecase Diagram



3-Non Functional Requirements

3.1- Performance Requirements

- All database queries must complete within 2 seconds under normal load (single user, local MySQL).
- The system must handle a minimum of 50 concurrent equipment records and 200 booking records without performance degradation.
- Application startup time must not exceed 5 seconds on a standard development machine.

3.2 -Safety Requirements

- The system must prevent double-booking of equipment by validating time slot conflicts before creating a booking.
- Equipment with FAULTY or UNDER_REPAIR status must not be bookable by teachers.
- Deletion of equipment is not permitted if active bookings exist.

3.3- Security Requirements

- Passwords are hashed using SHA-256 via PasswordUtil before storage.
- Role-based access control: Teachers cannot access Lab Manager or Technician functions.
- All significant actions (login, booking, status changes) are recorded in the audit_log table.
- SQL queries use parameterized PreparedStatements to prevent SQL injection.

3.4- Software Quality Attributes

- Maintainability: Code is divided into distinct layers (UI, Service, DAO) with low coupling and high cohesion.
- Reliability: The Singleton DatabaseConnection ensures a consistent, single connection point with isAlive() health checking.
- Testability: Each service class is independently testable with mock DAO objects.
- Portability: The application runs on any OS with Java 17+ and MySQL 8.0+.
- Usability: Console output is clearly labeled per use case with meaningful status messages.

3.5 -Business Rules

- Only users with role TEACHER can create bookings.
- Only users with role LAB_MANAGER can approve, reject, or assign maintenance tasks.
- Only users with role TECHNICIAN can update repair status and mark faults as resolved.
- A booking must have start time before end time.
- Equipment in RETIRED status cannot be booked or assigned a maintenance task.
- Each fault report must be assigned to exactly one technician before repair can begin.

3.6 -Operating Environment

Component	Specification
Programming Language	Java 17 (LTS)

Build Tool	Apache Maven 3.8+
Database	Microsoft SQL Server 2017+ Express Edition
JDBC Driver	mssql-jdbc 12.x(Microsoft SQL Server JDBC Driver)
Operating System	Windows 10/11, macOS, Linux (any with JDK 17)
RAM	Minimum 4 GB
IDE	Visual Studio Code (with extension pack of Java)

3.7 User Interfaces

The current implementation uses a console-based UI (Main.java) that auto-executes all 15 use cases sequentially and prints results to standard output. The UI demonstrates the complete workflow across Teacher, Lab Manager, and Technician roles.

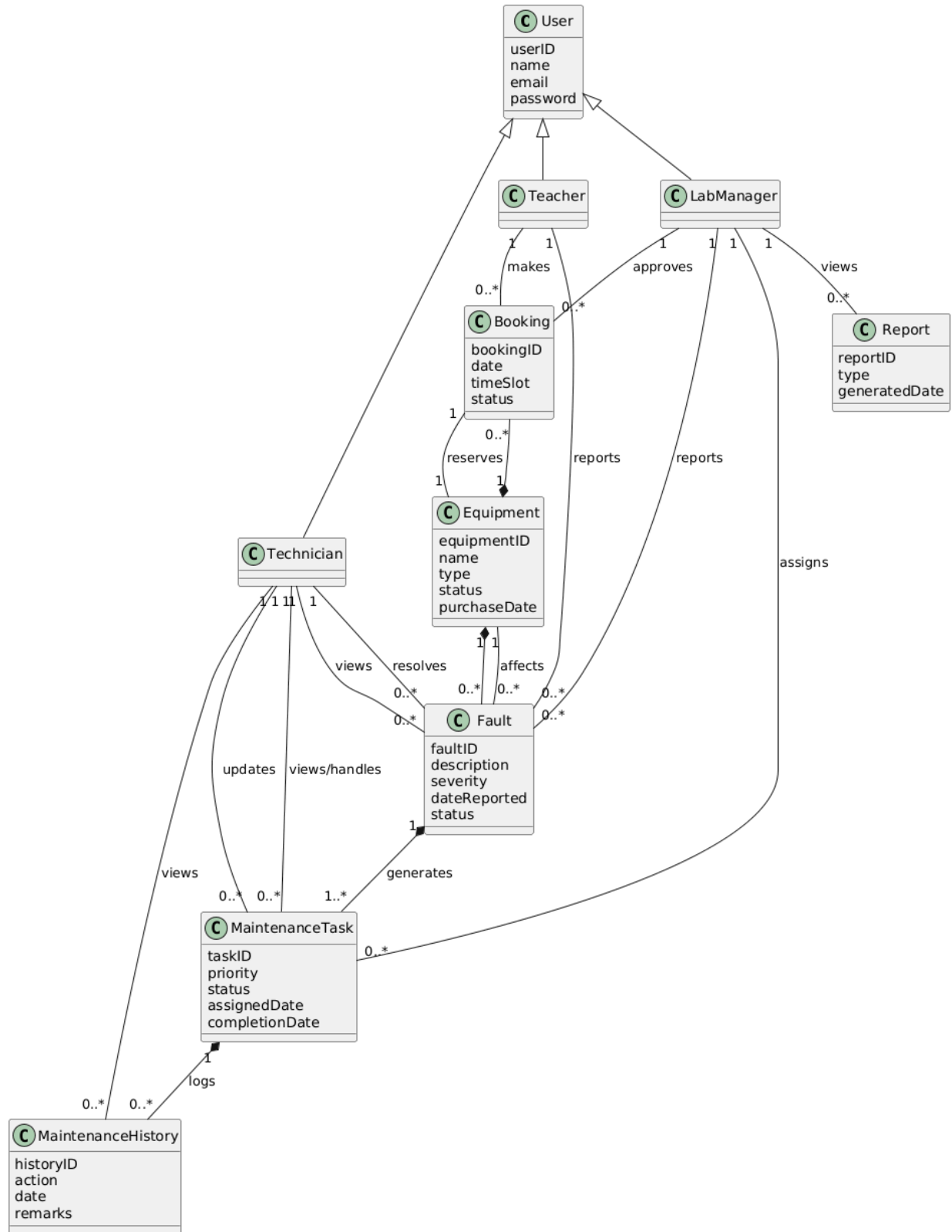
Console output format example:

[UC03-Teacher] Book Equipment

Booking created: Booking{id=7, status=PENDING, ...}

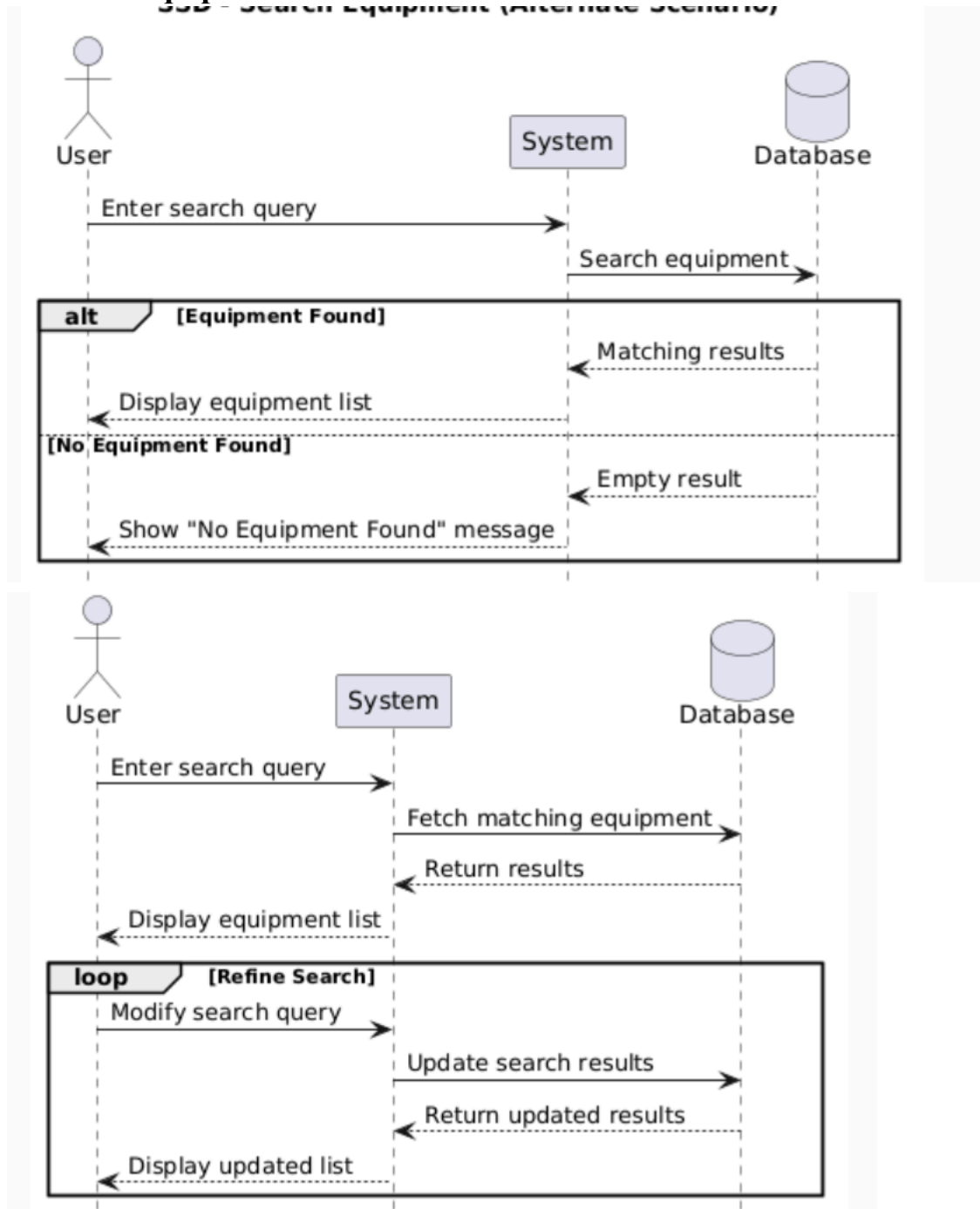
A future enhancement would replace the console with a JavaFX GUI with separate screens for each role.

4-Domain Model

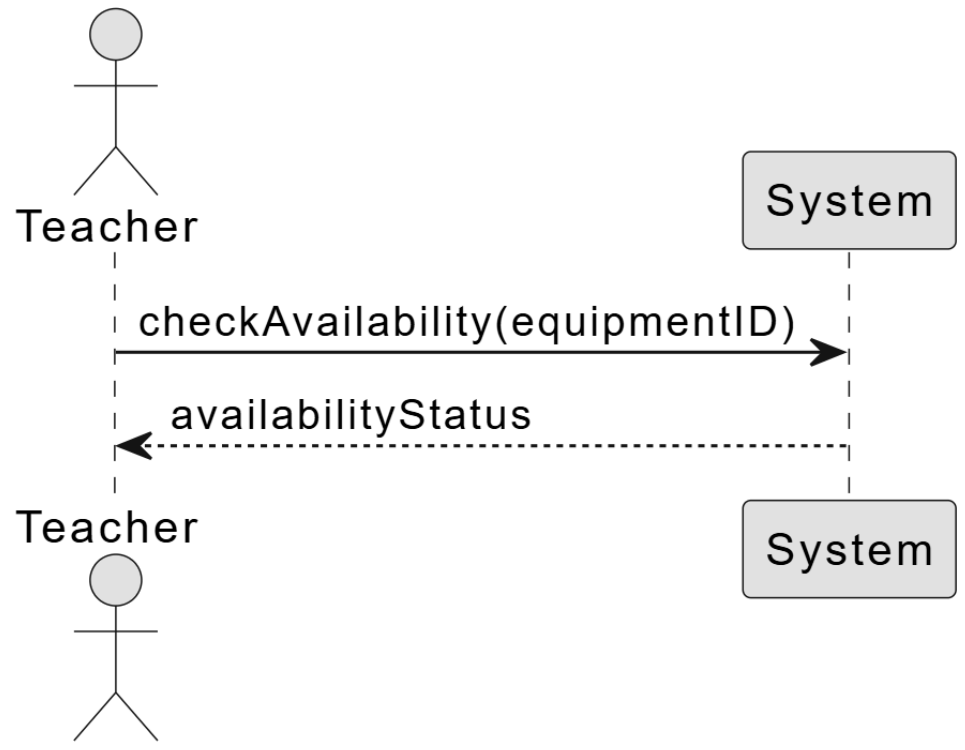


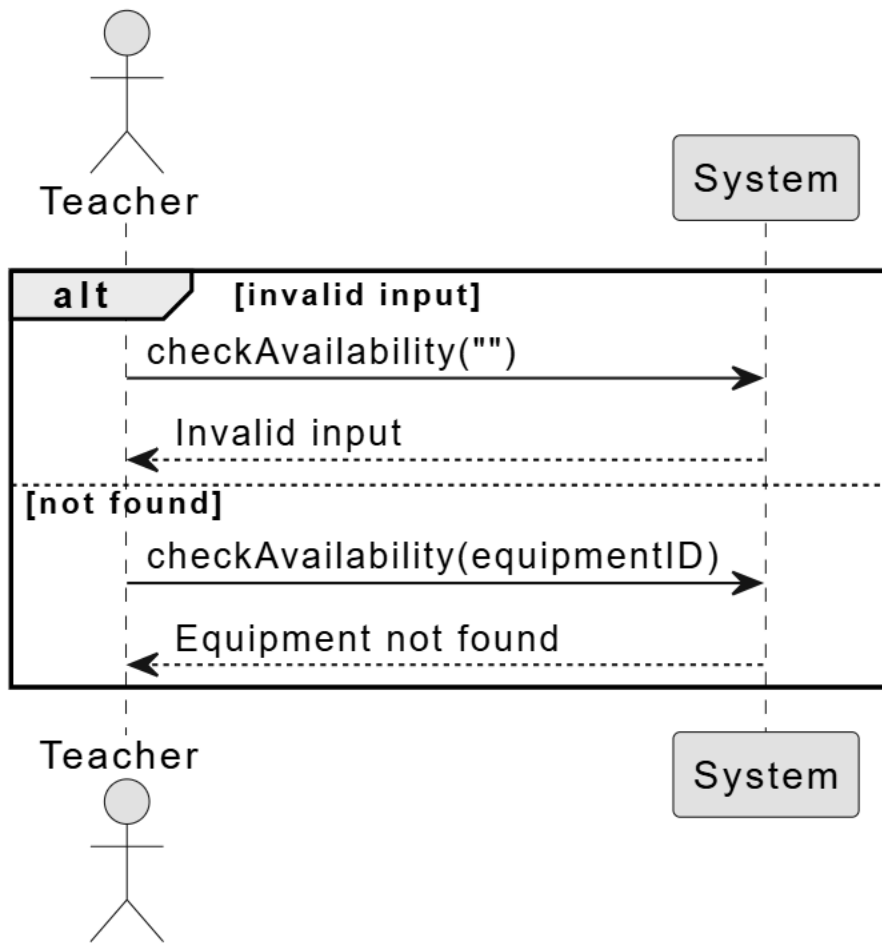
5-System Sequence Diagrams

1-Search Equipment

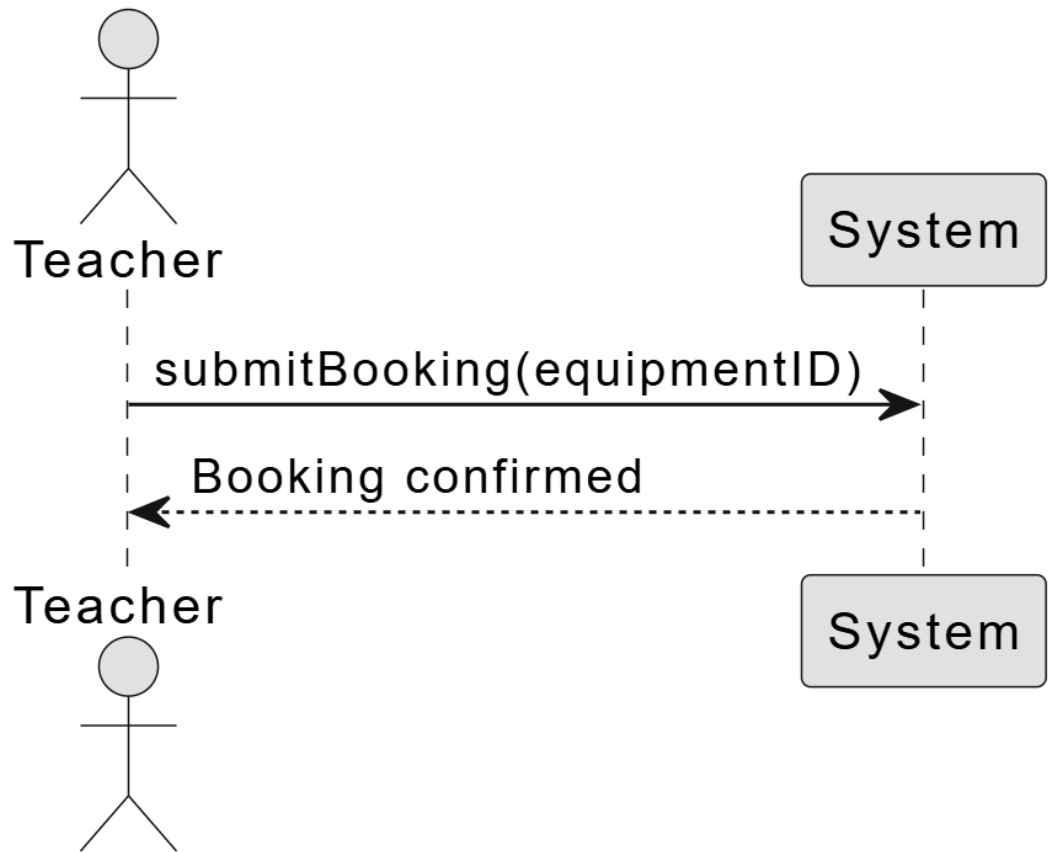


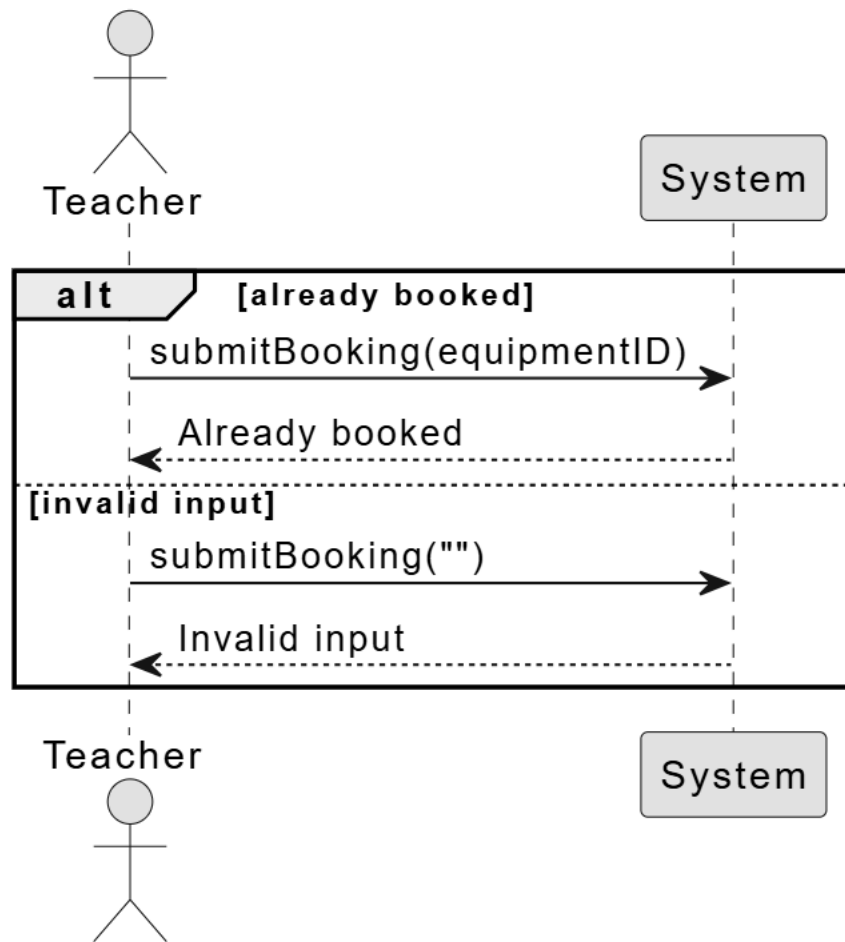
2- Check Equipment Availability



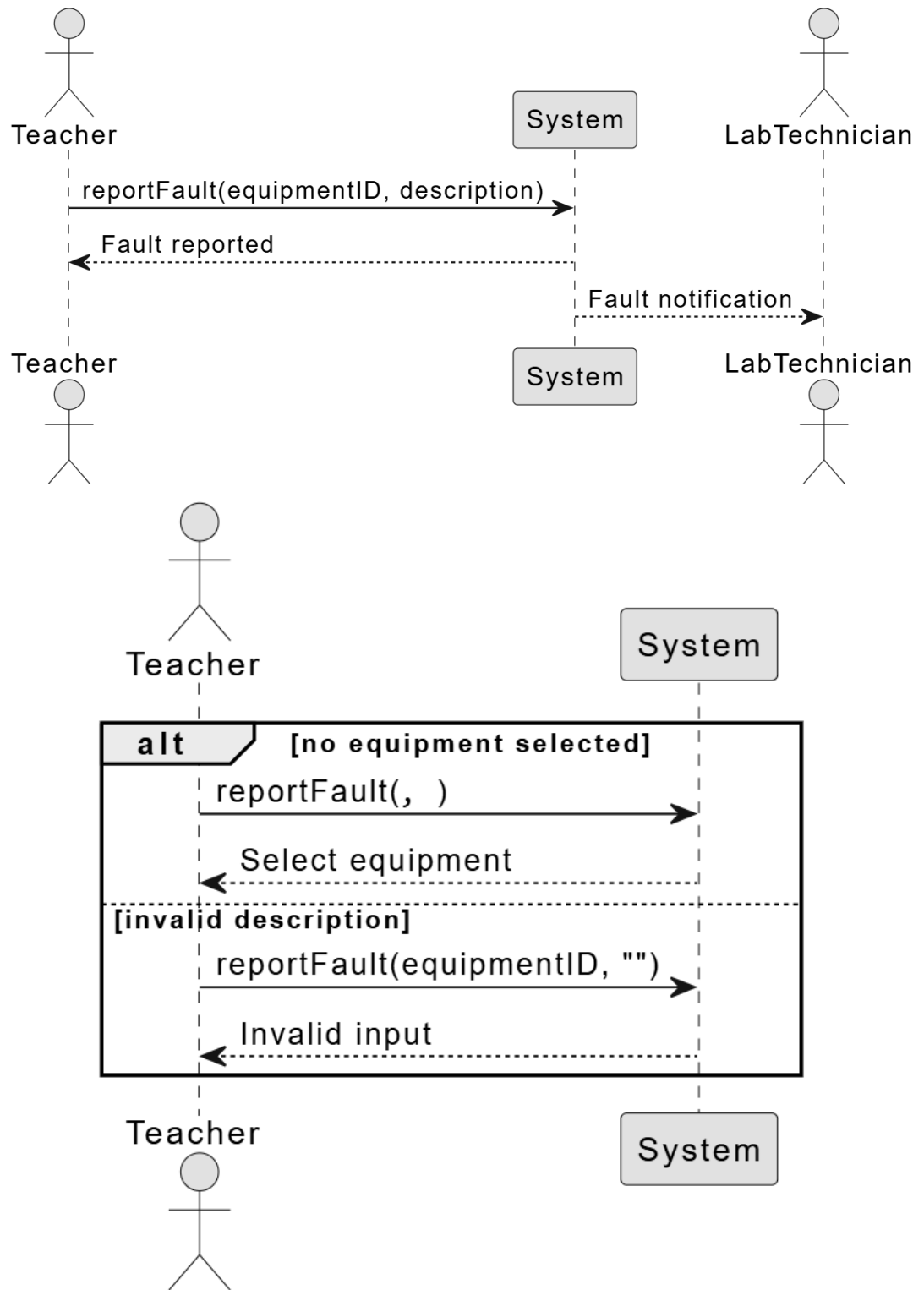


3- Booking Equipment

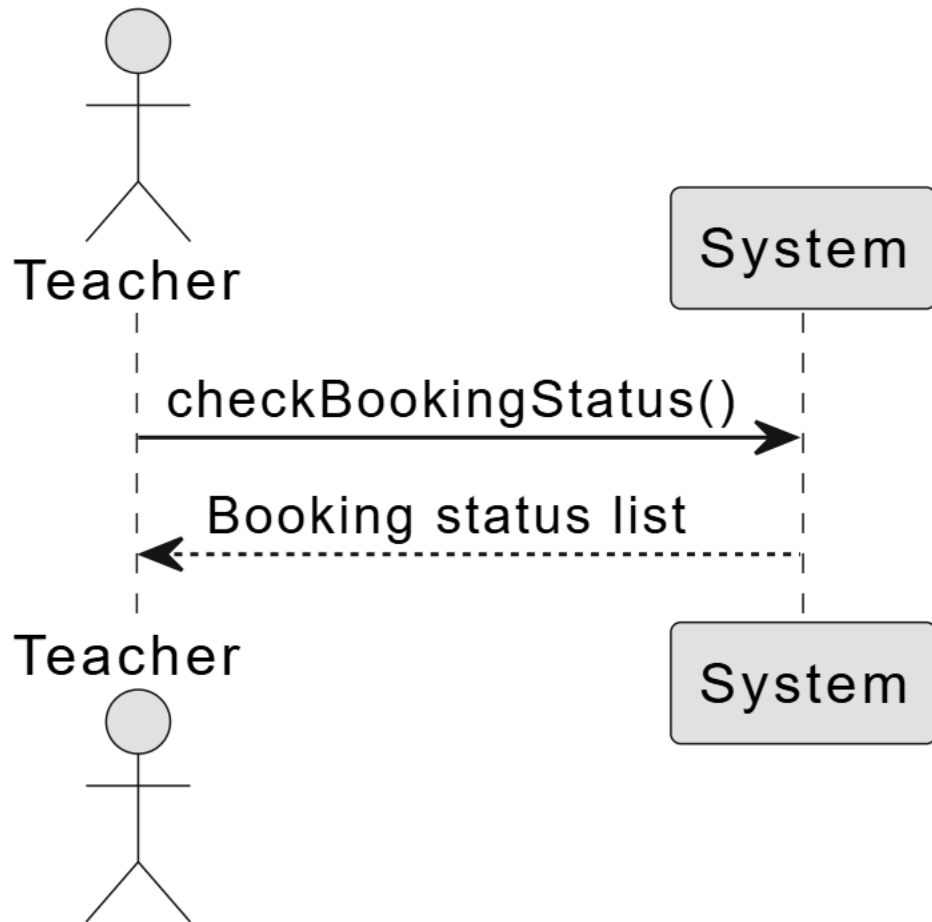


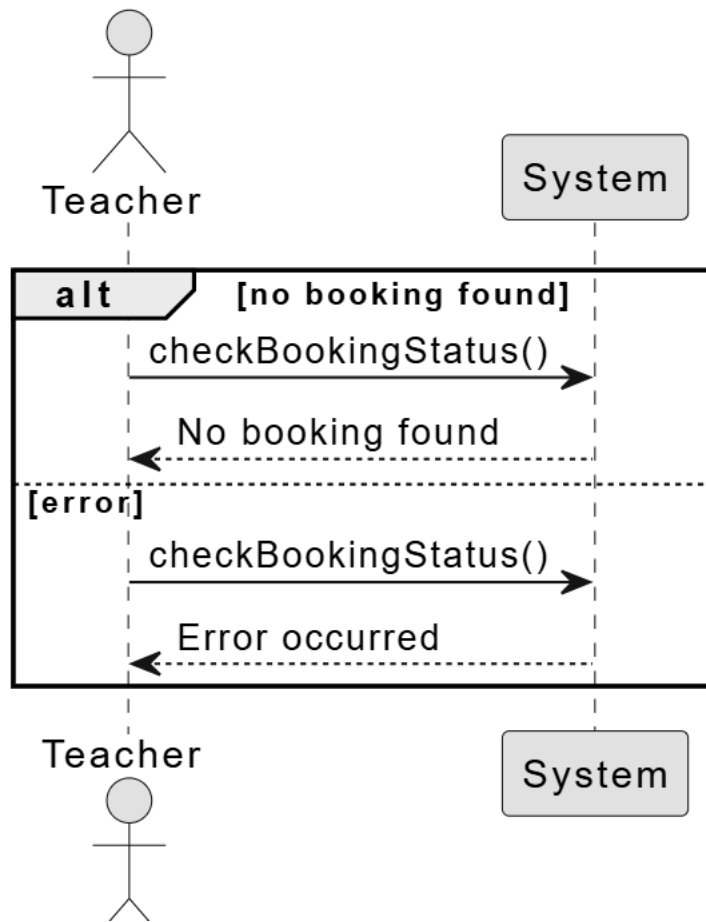


4- Report Faulty Equipment



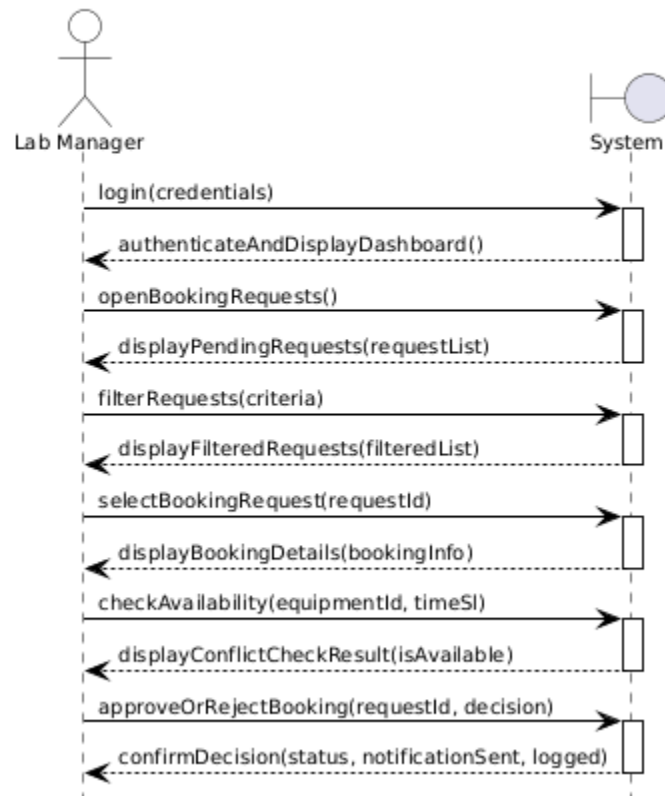
5-Check Booking Status



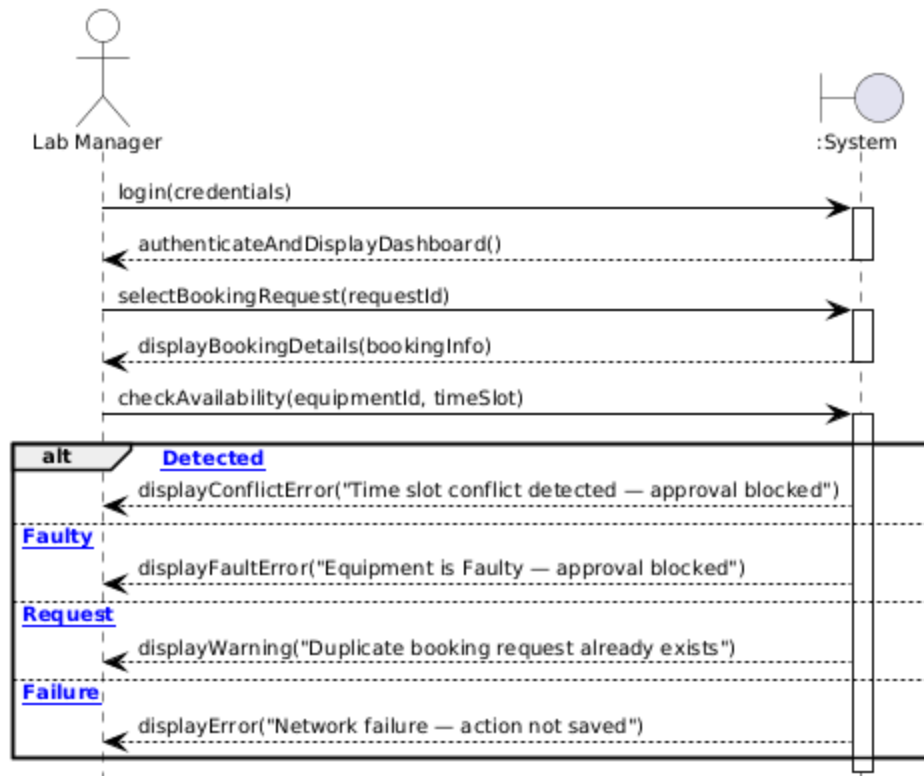


6- Approve/Reject Booking

UC01 - Approve / Reject Booking : Main Success Scenario

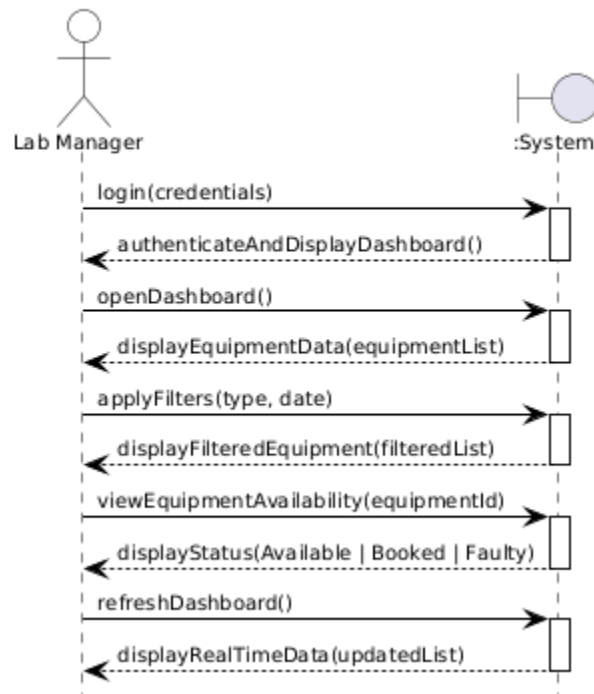


UC01 - Approve / Reject Booking : Alternative Scenarios

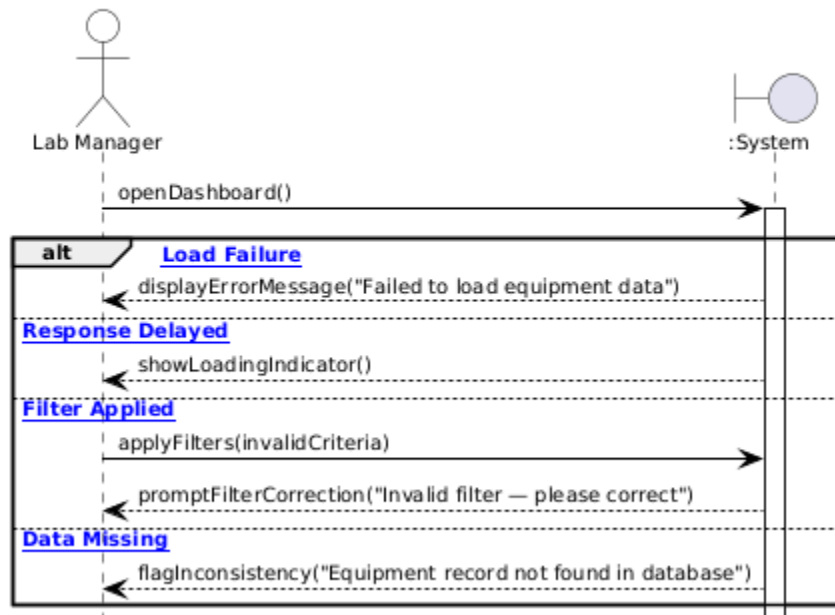


7-Monitor Equipment Availability

UC02 - Monitor Equipment Availability : Main Success Scenario

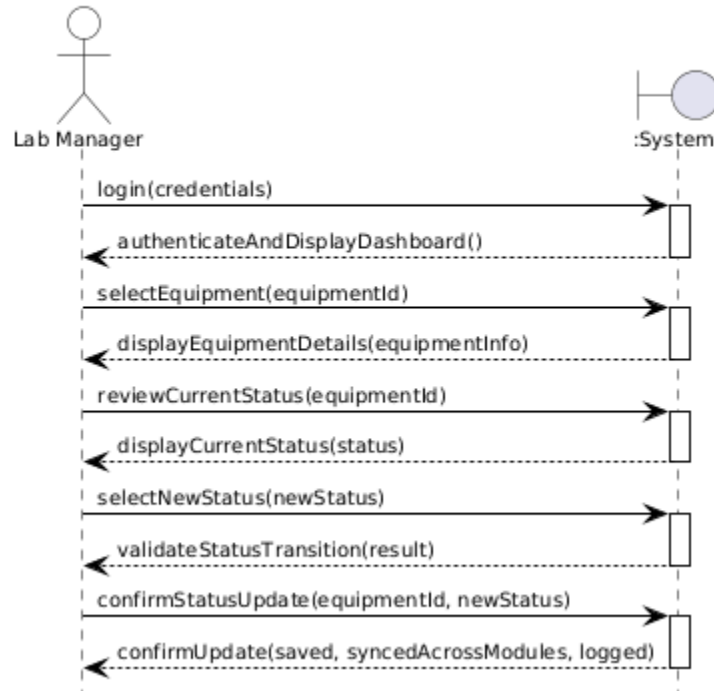


UC02 - Monitor Equipment Availability : Alternative Scenarios

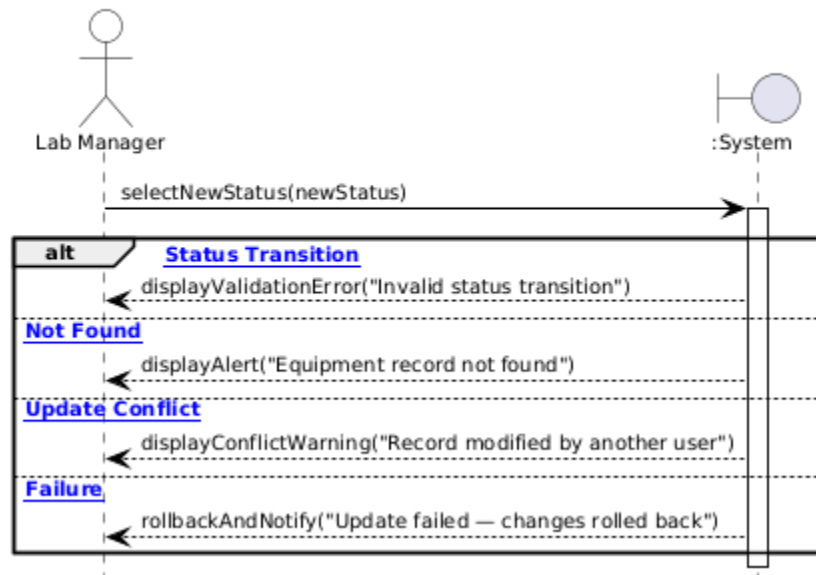


8- Update Equipment Status

UC03 - Update Equipment Status : Main Success Scenario

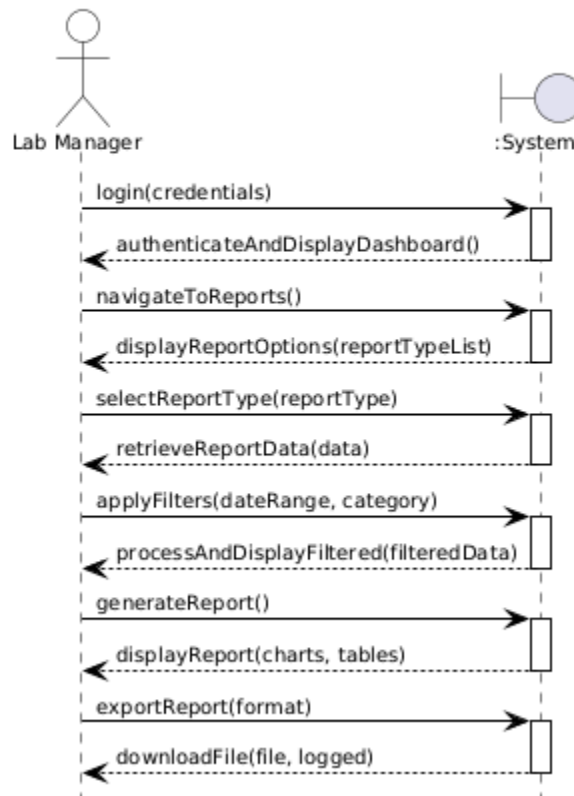


UC03 - Update Equipment Status : Alternative Scenarios

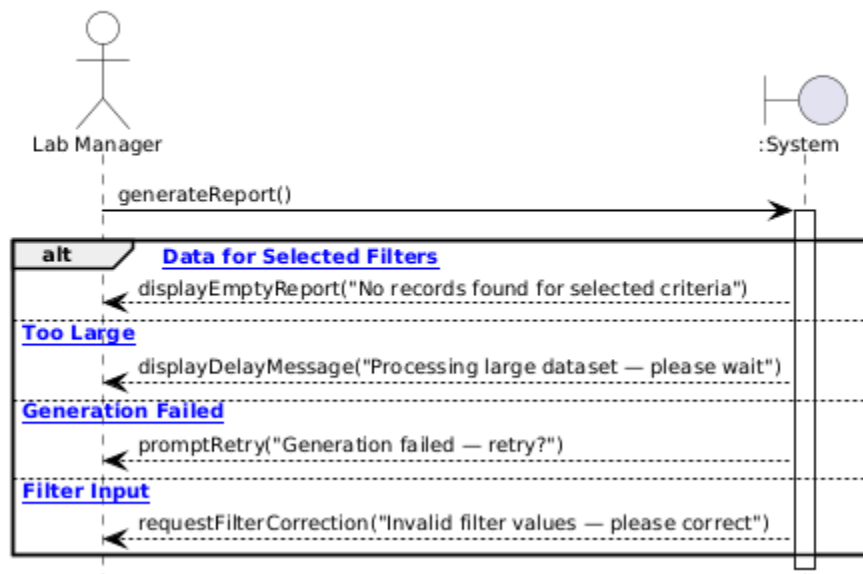


9-View Reports and Availability

UC04 - View Reports and Analytics : Main Success Scenario

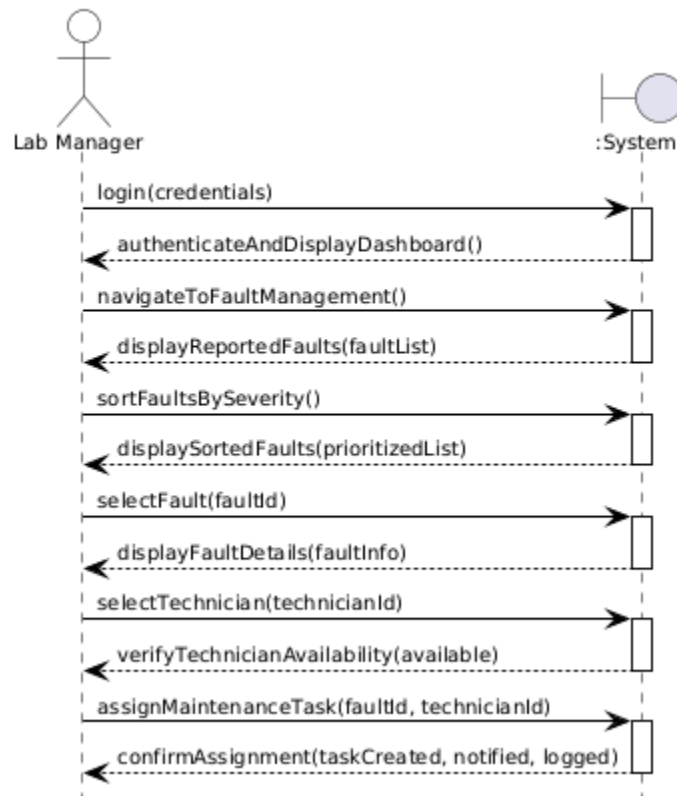


UC04 - View Reports and Analytics : Alternative Scenarios

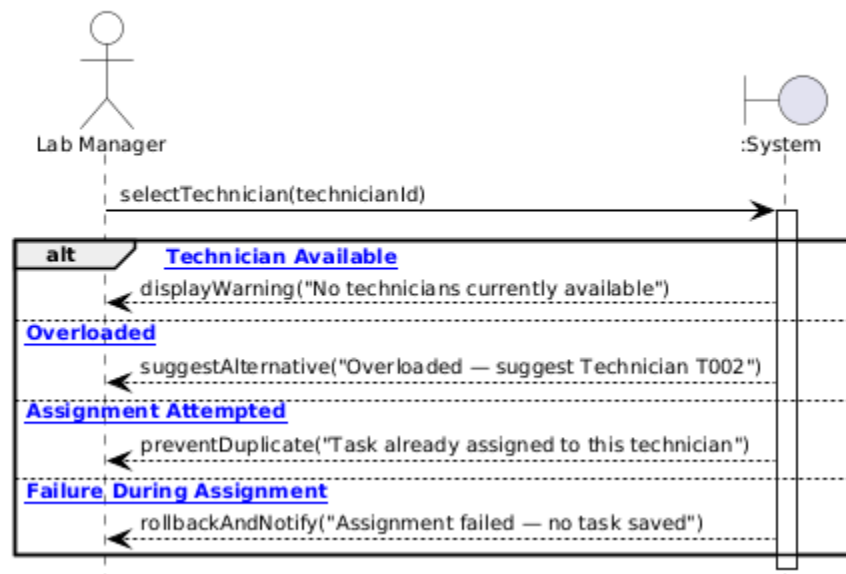


10-Assign Maintenance Tasks

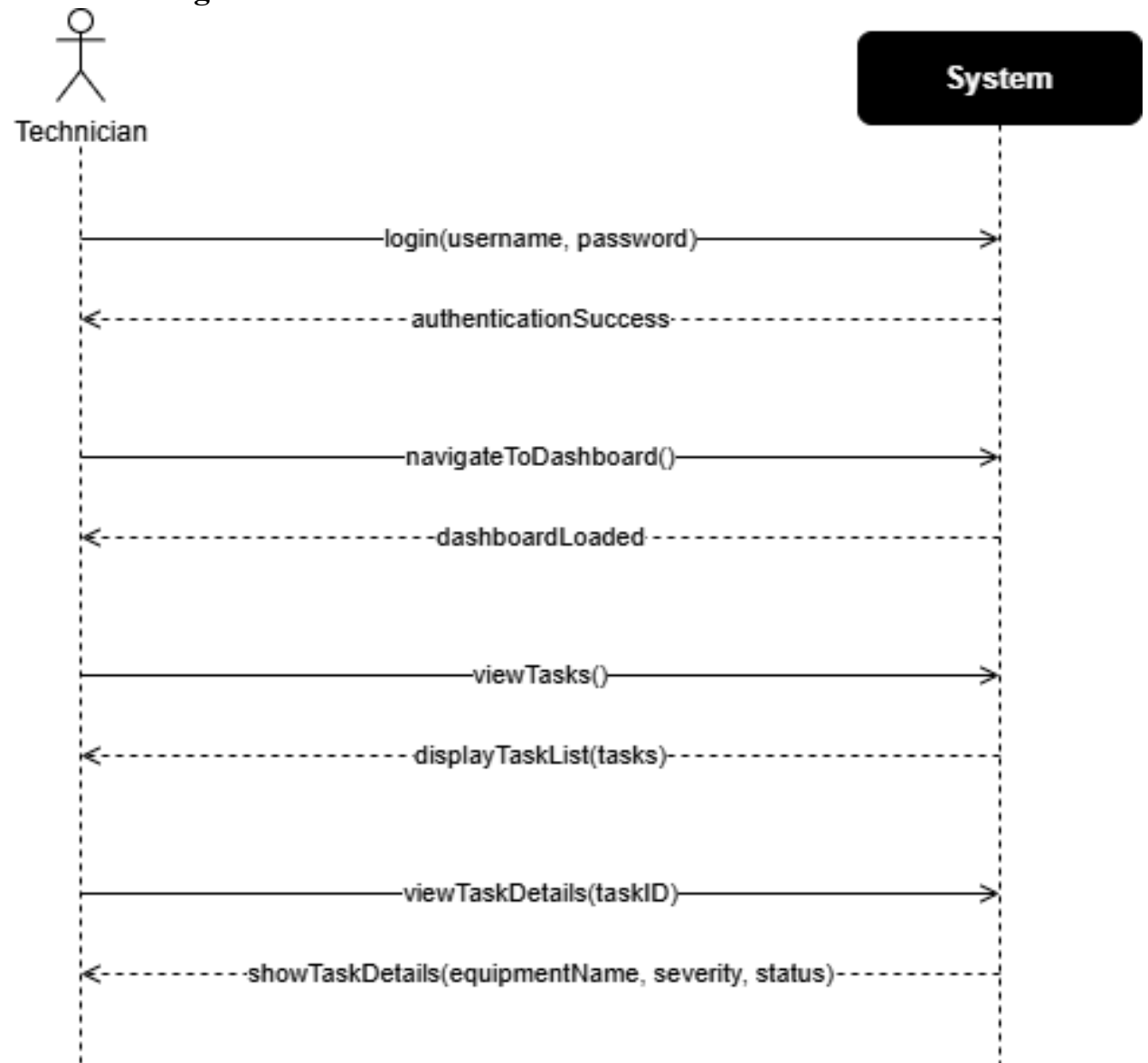
UC05 - Assign Maintenance Tasks : Main Success Scenario

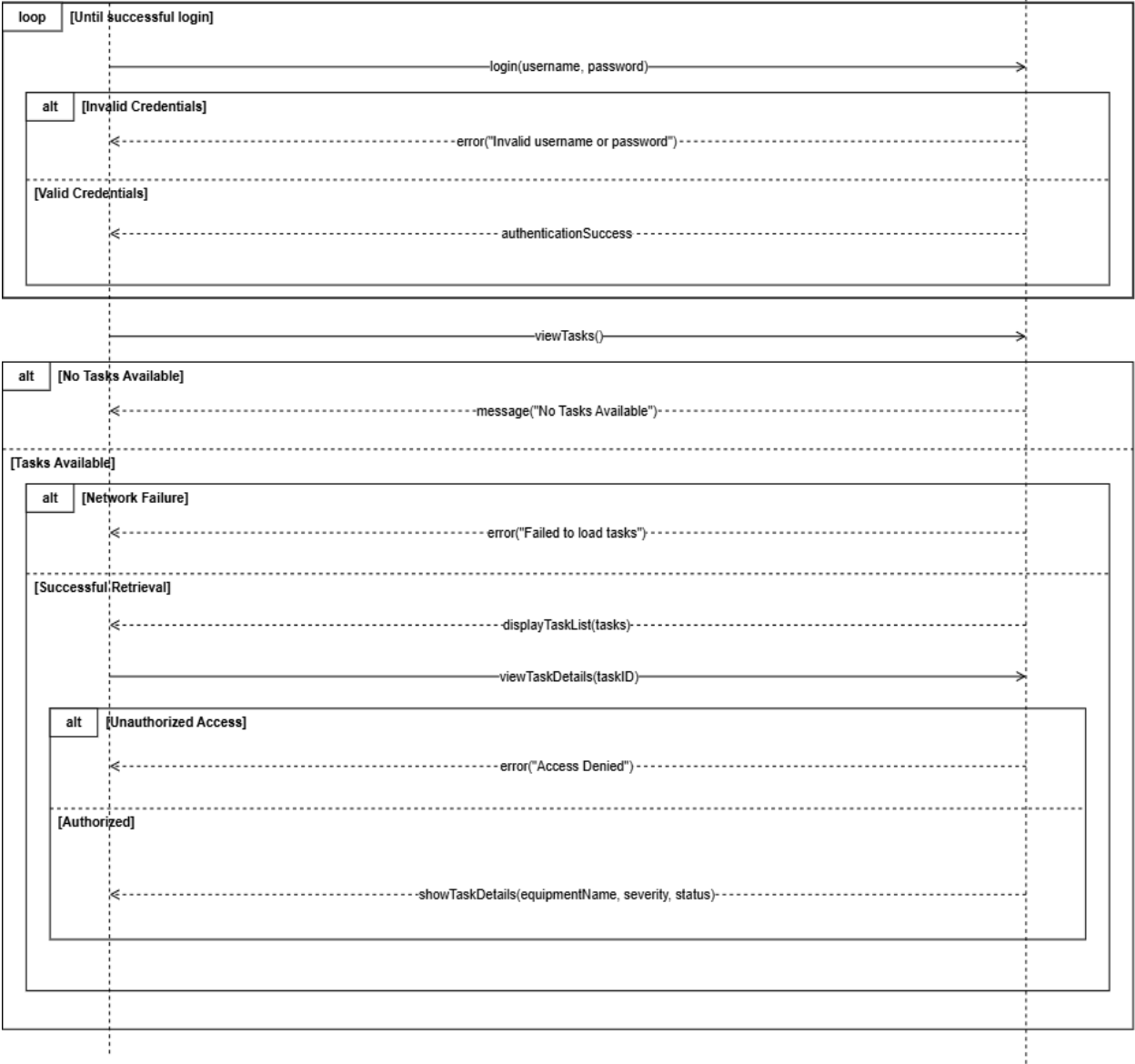


UC05 - Assign Maintenance Tasks : Alternative Scenarios

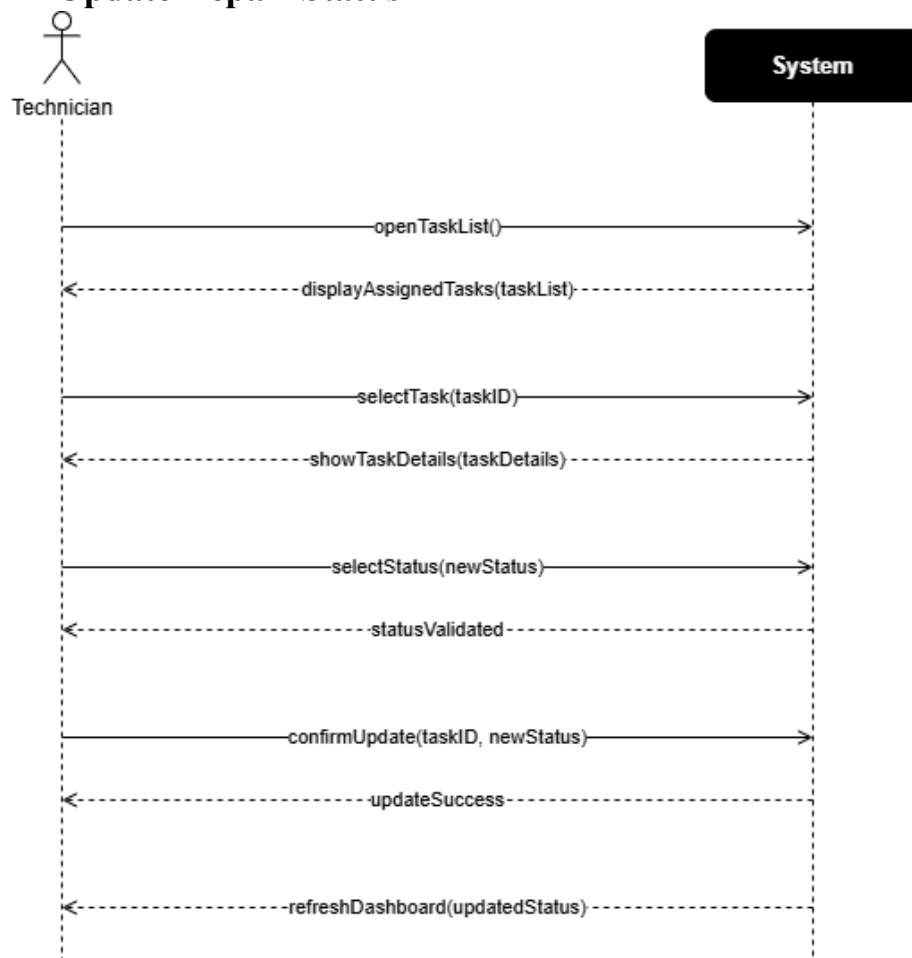


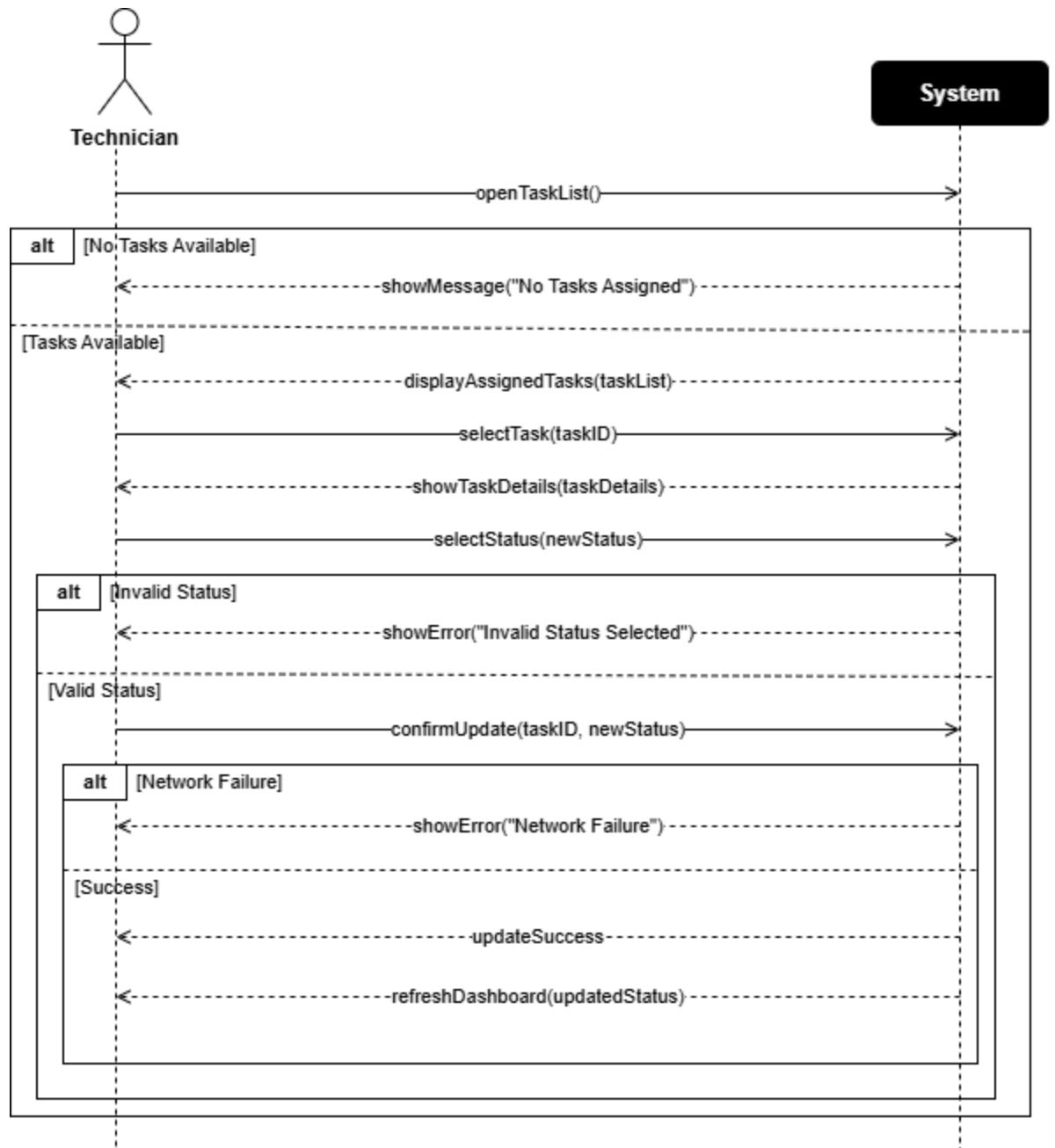
11-View Assigned Maintenance Task



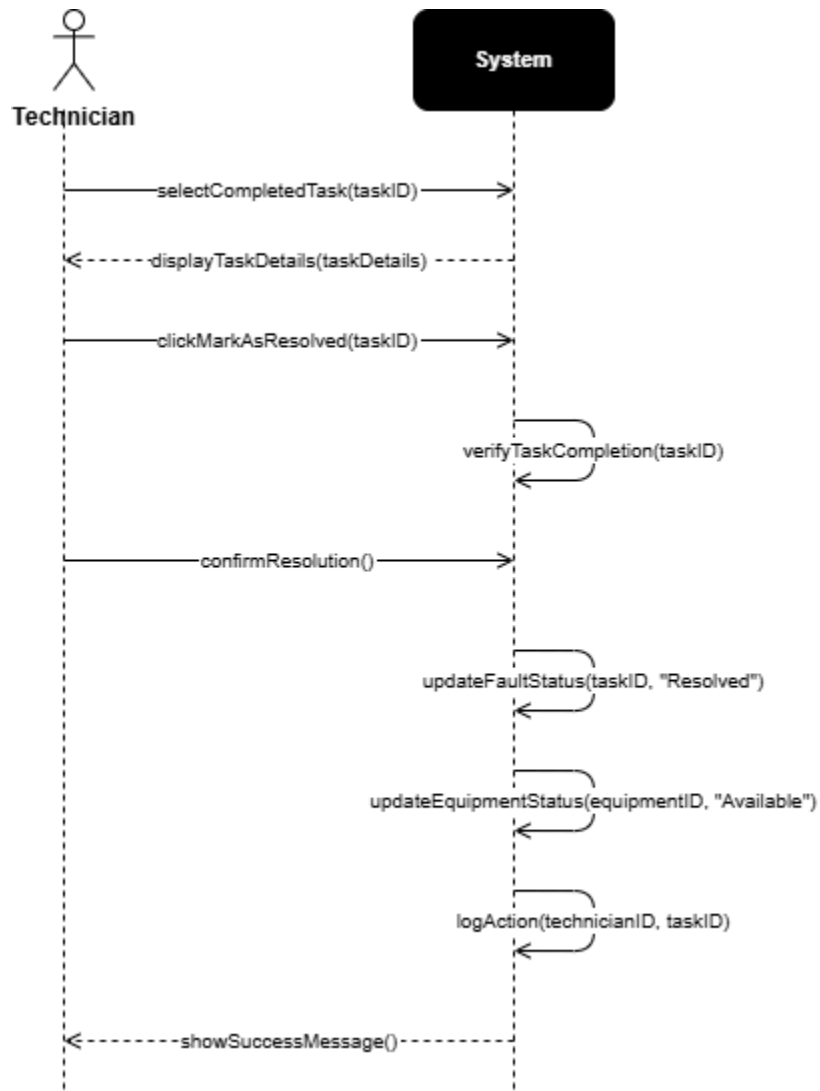


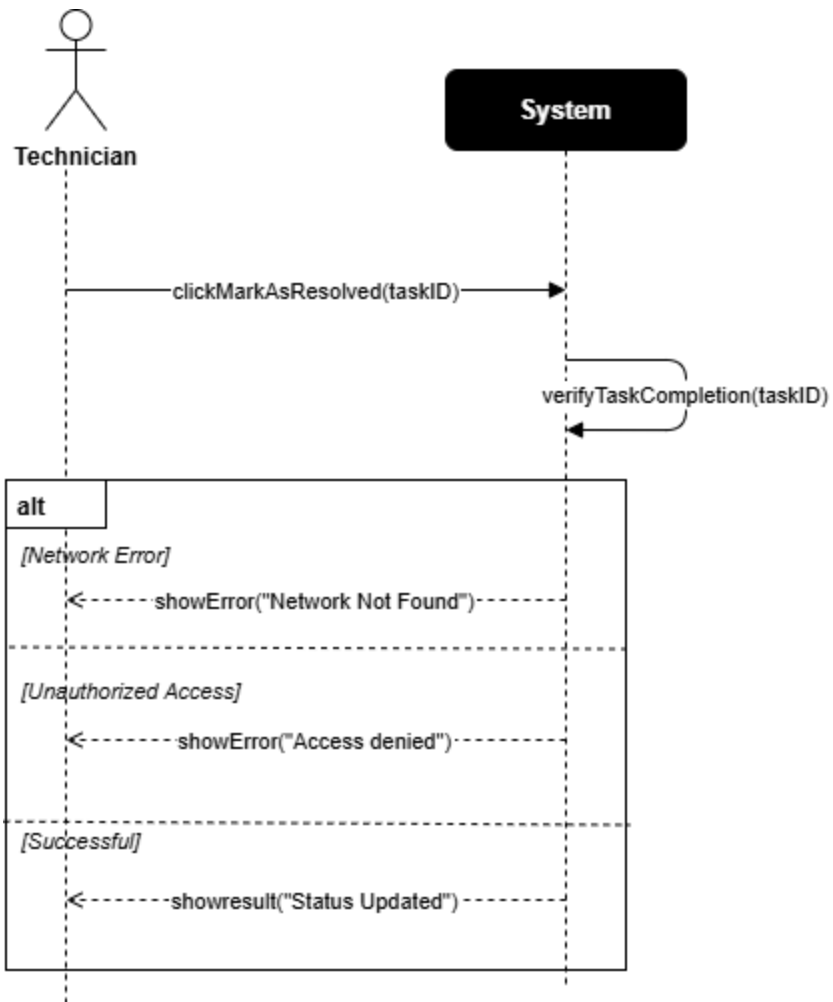
12-Update Repair Status



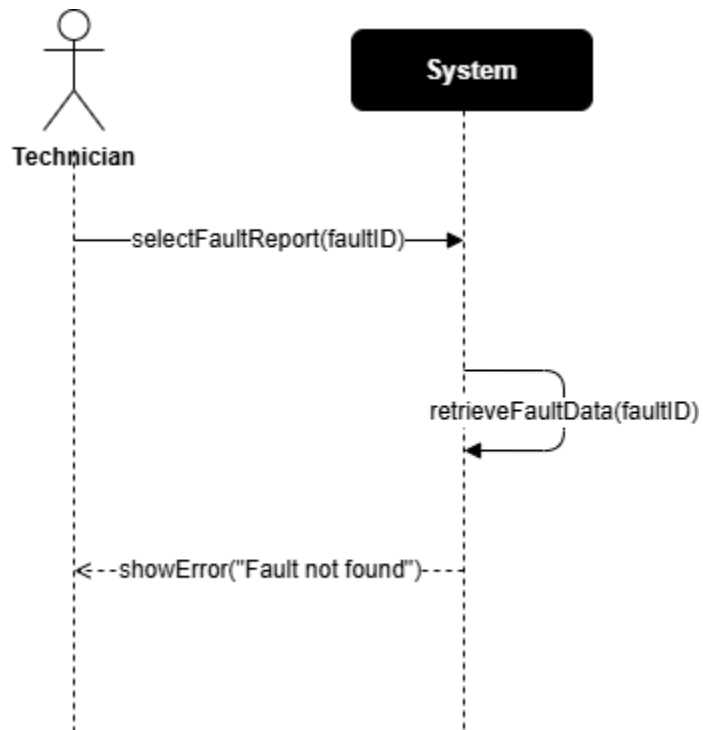
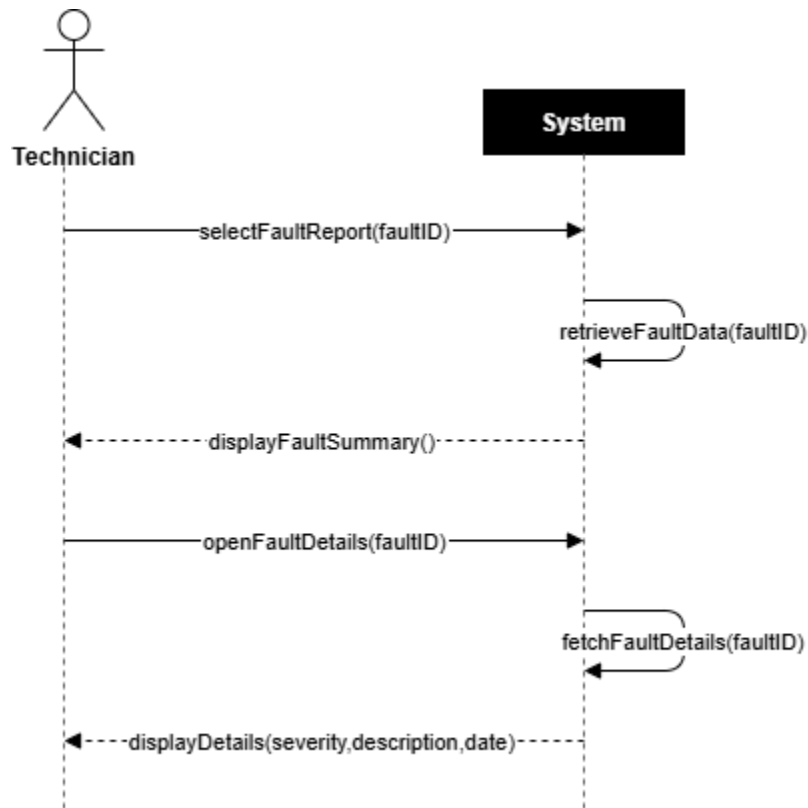


13- Marked Fault as Resolved

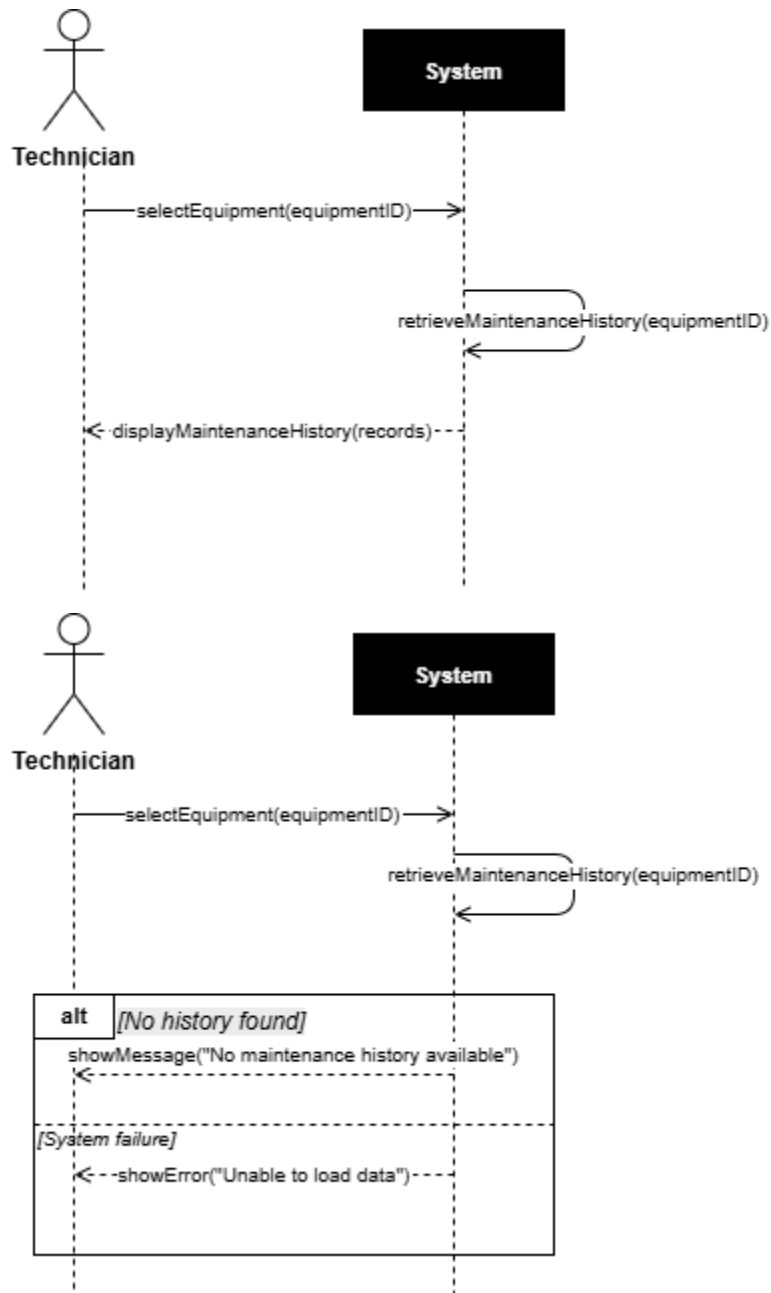




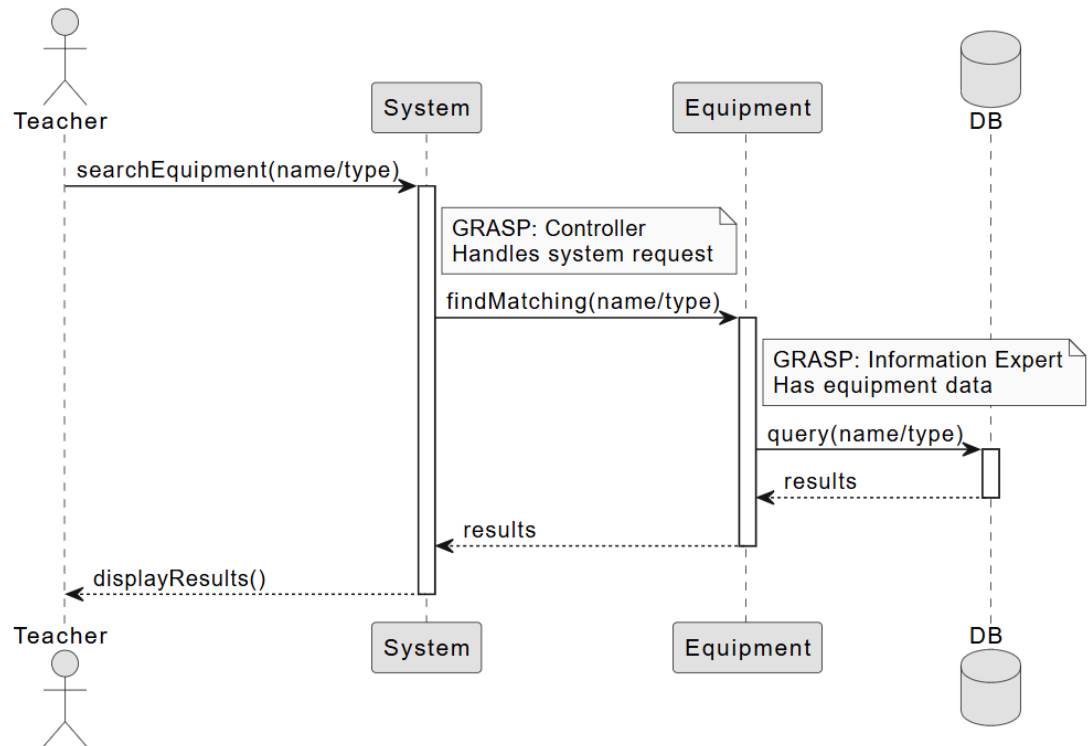
14-View Fault Details

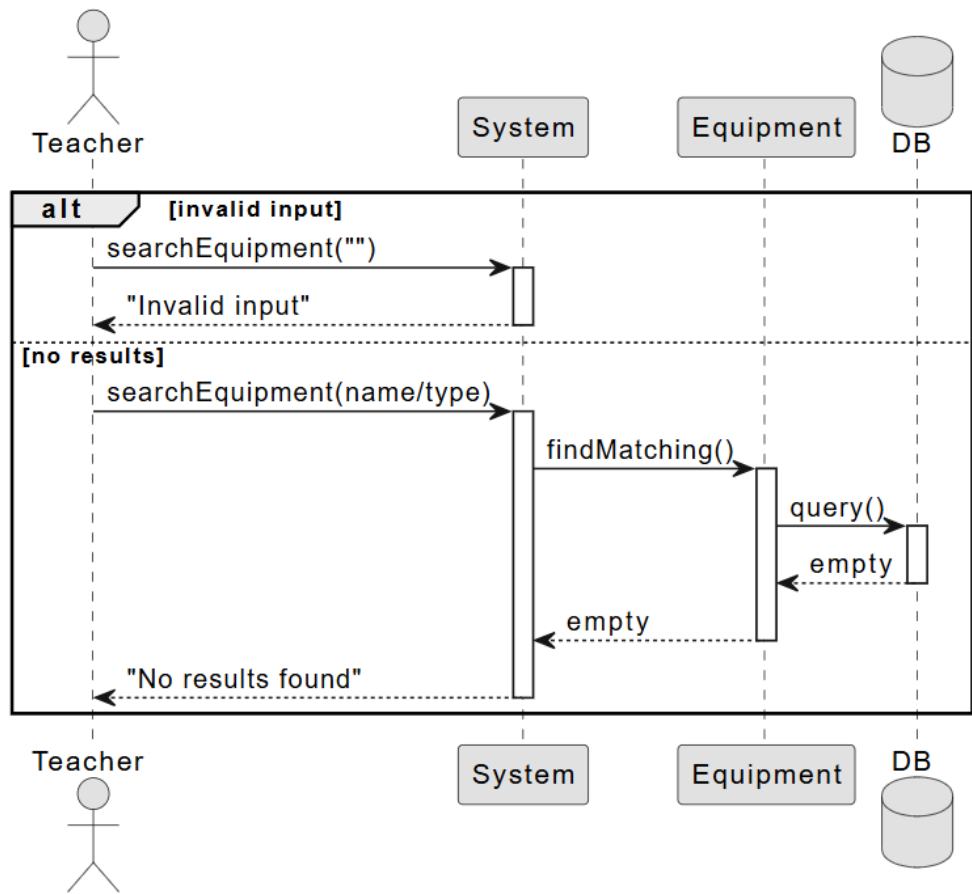


15-View Maintenance History

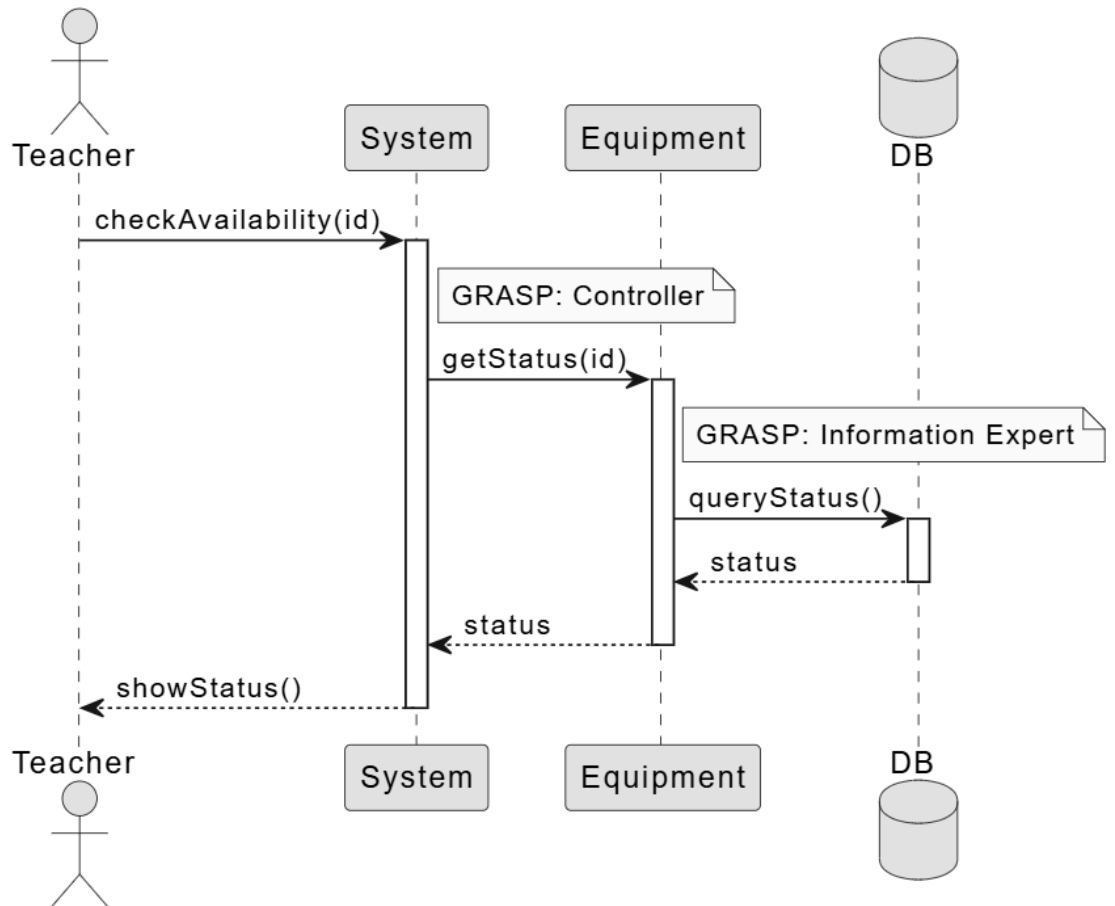


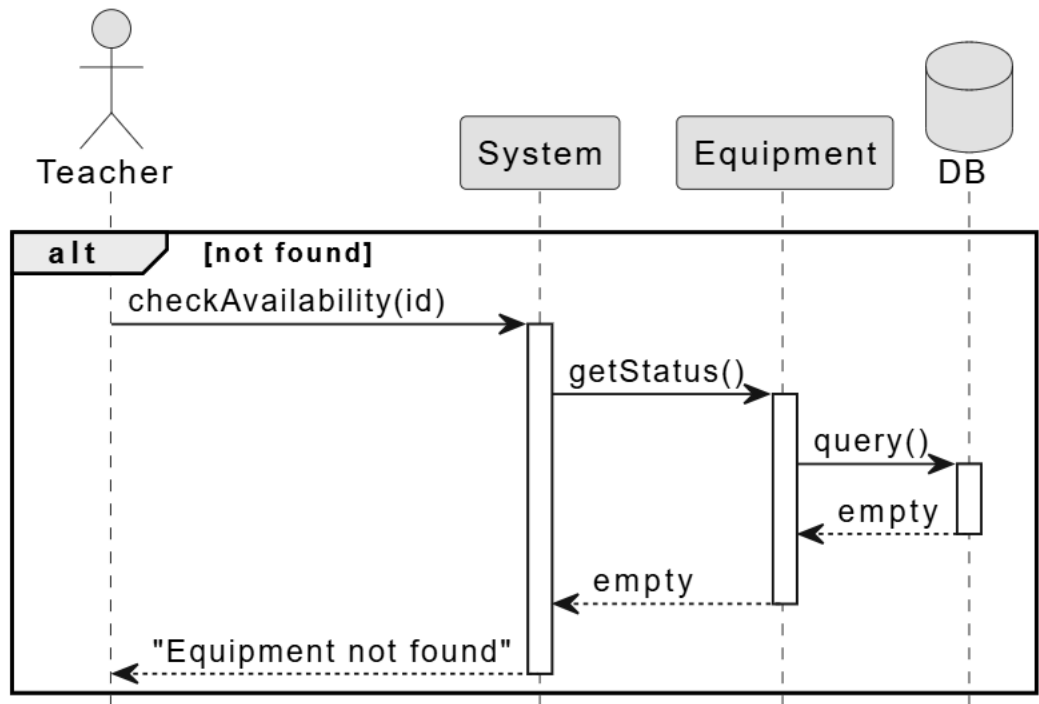
6-Sequence Diagram_ 1- Search Equipment



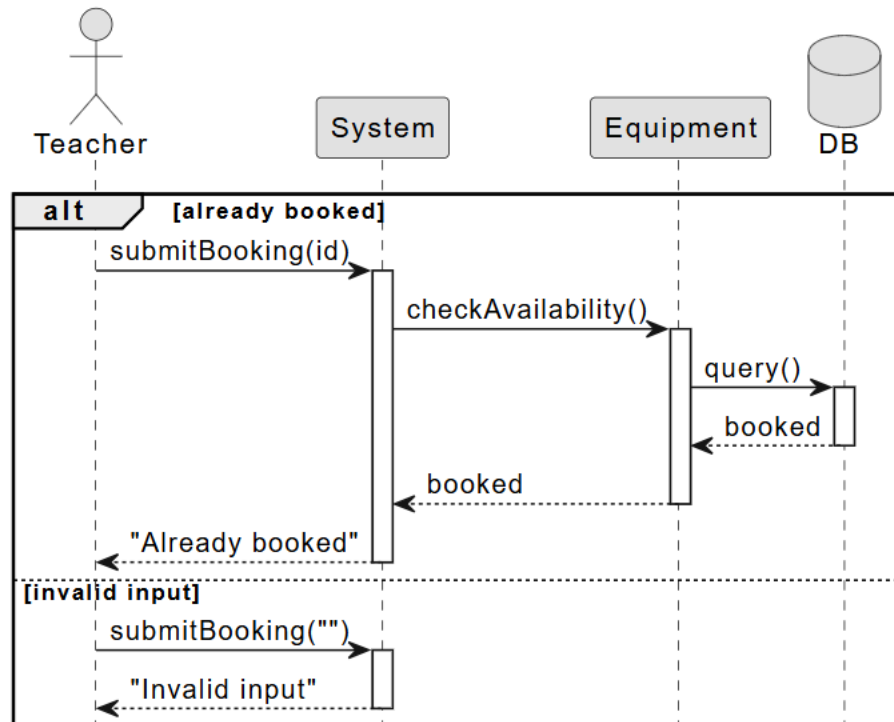
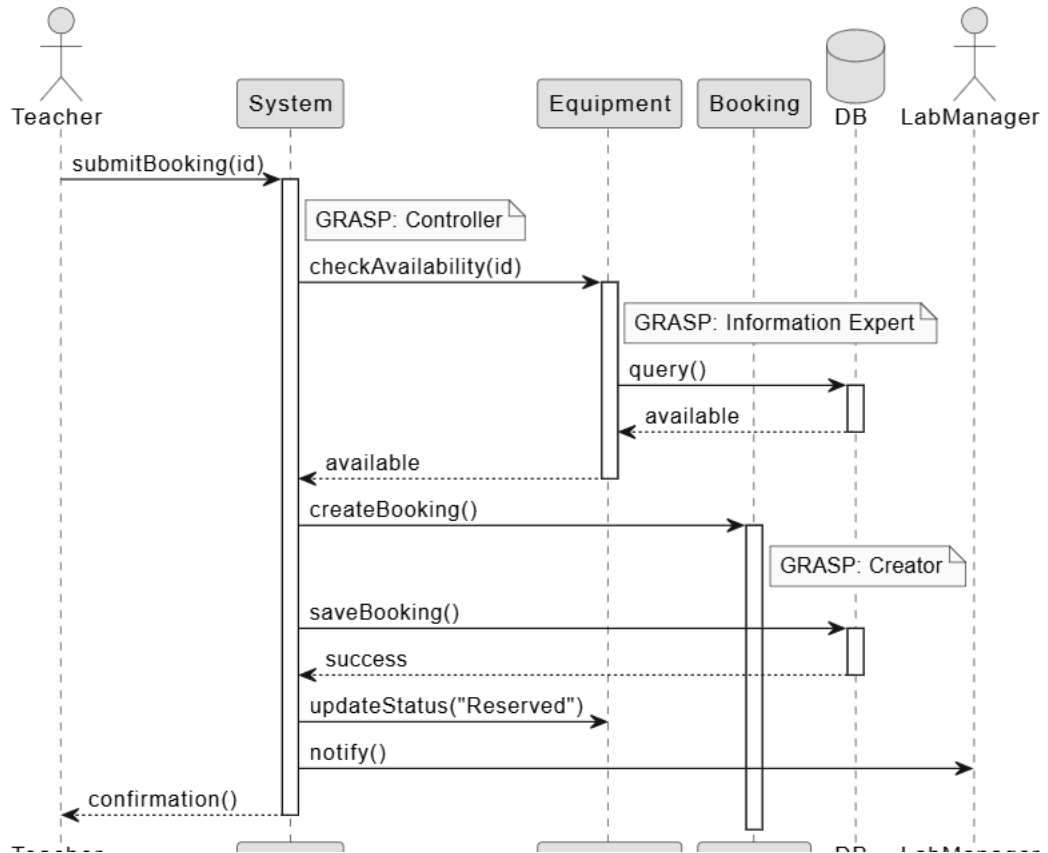


2-Check Equipment Availability

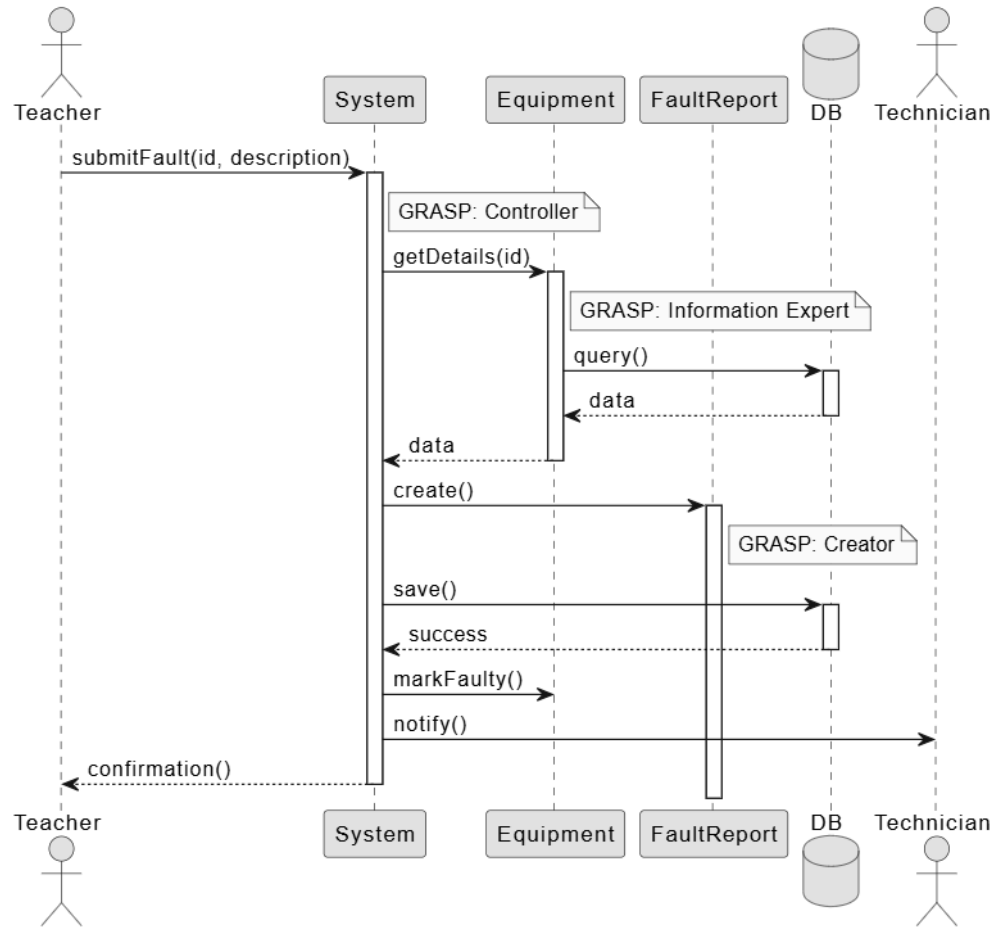




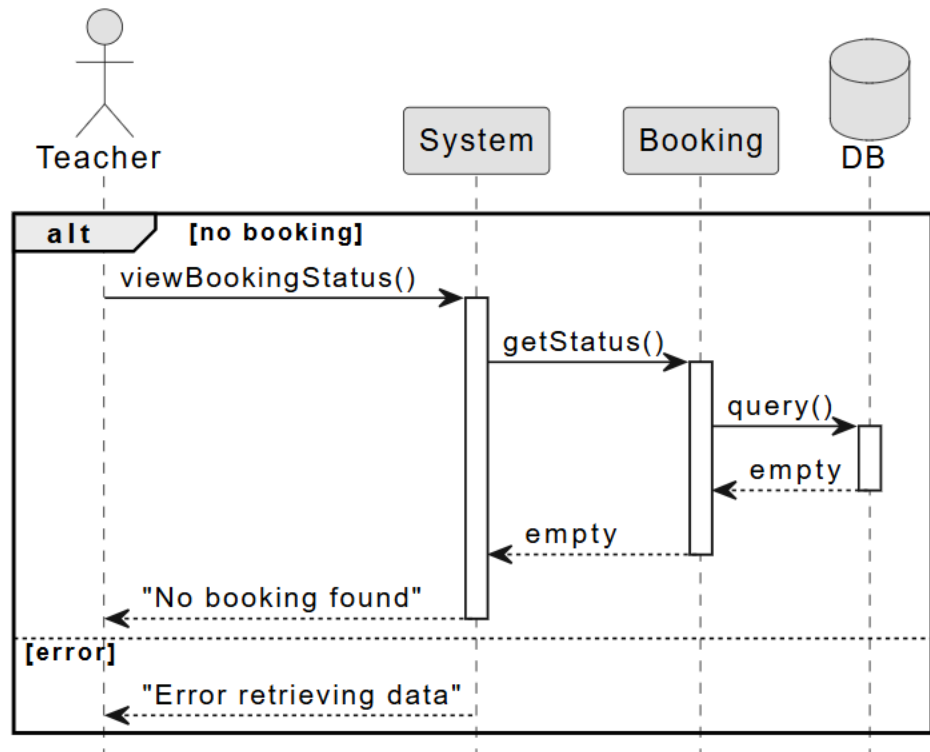
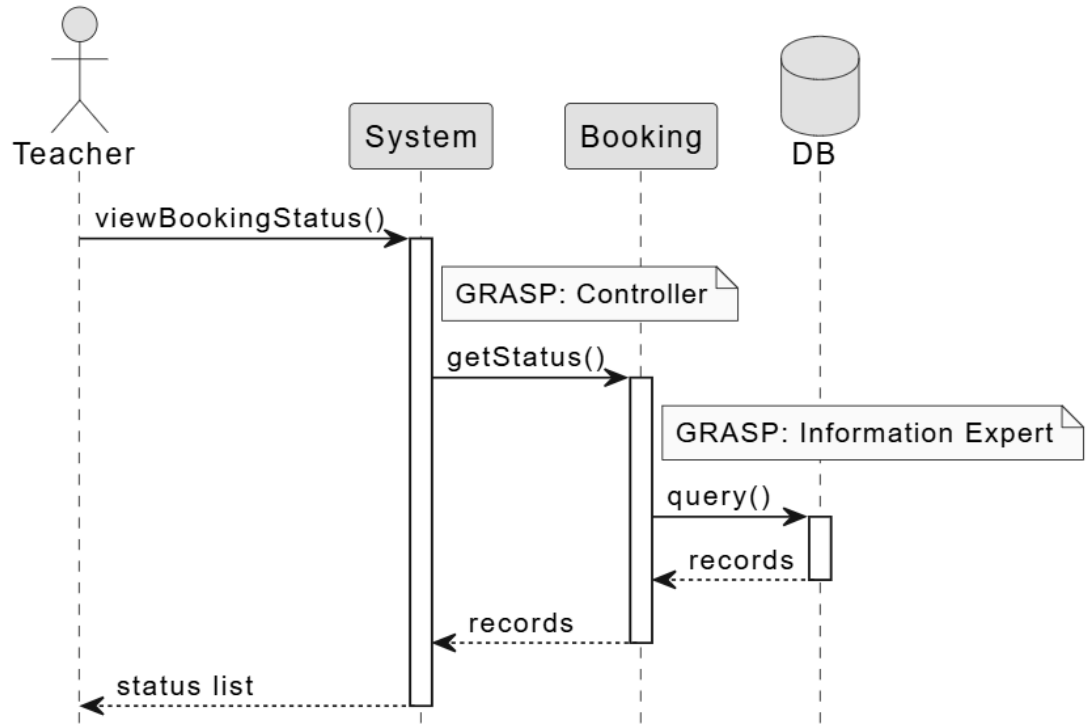
3-Book Equipment



4-Report Faulty Equipments

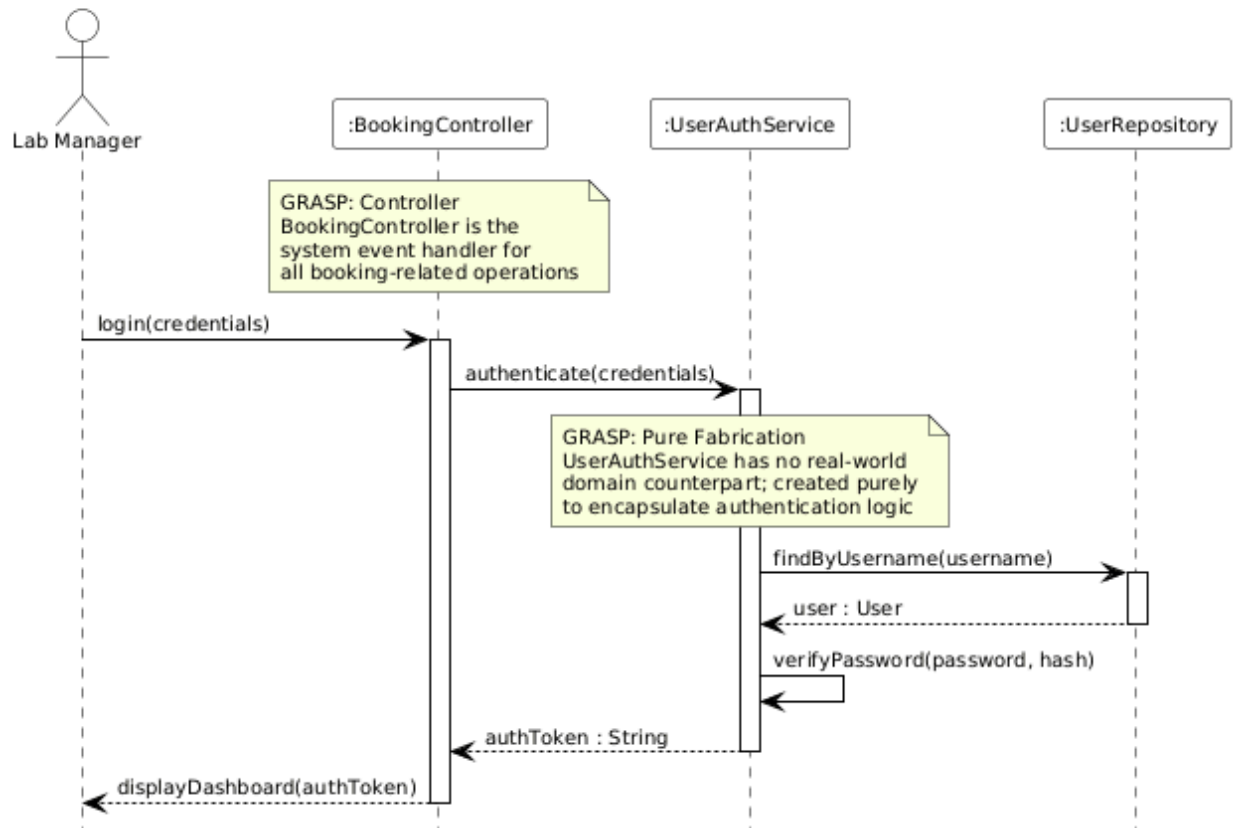


5-Check Booking Status

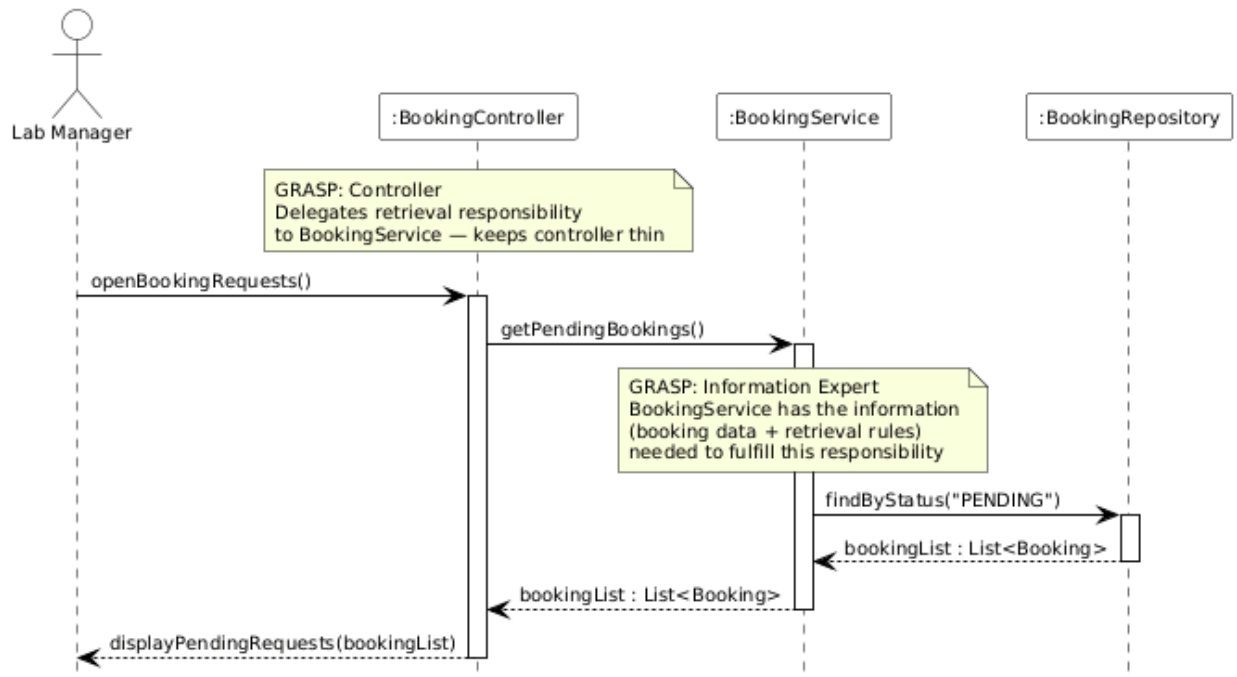


6- Approve/Reject Booking

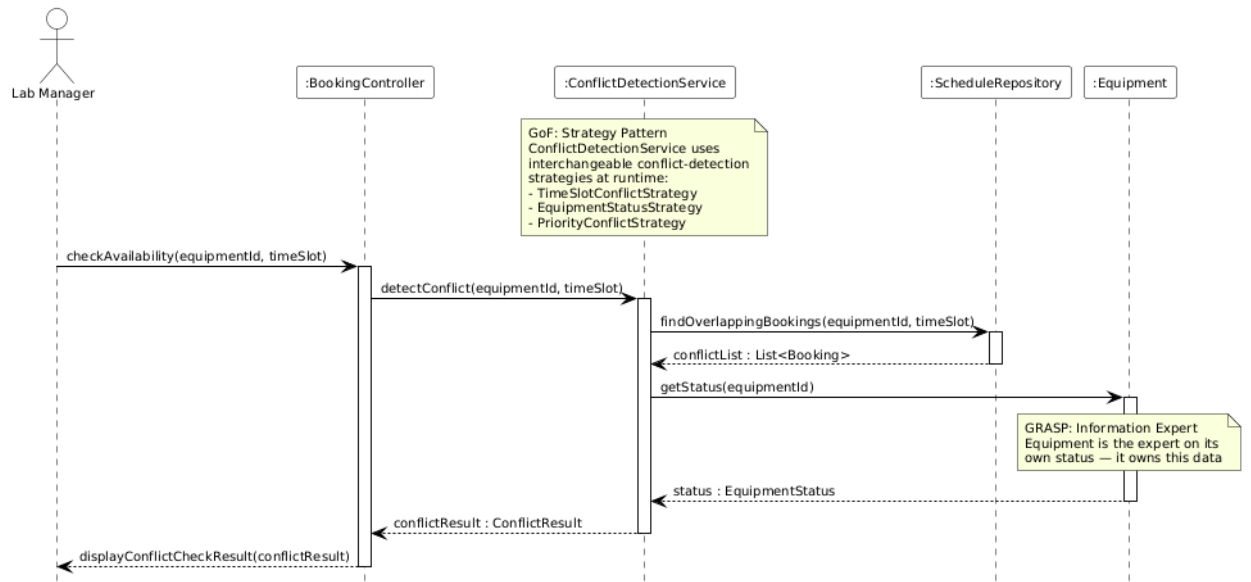
SD - UC01 Op1 : login(credentials)



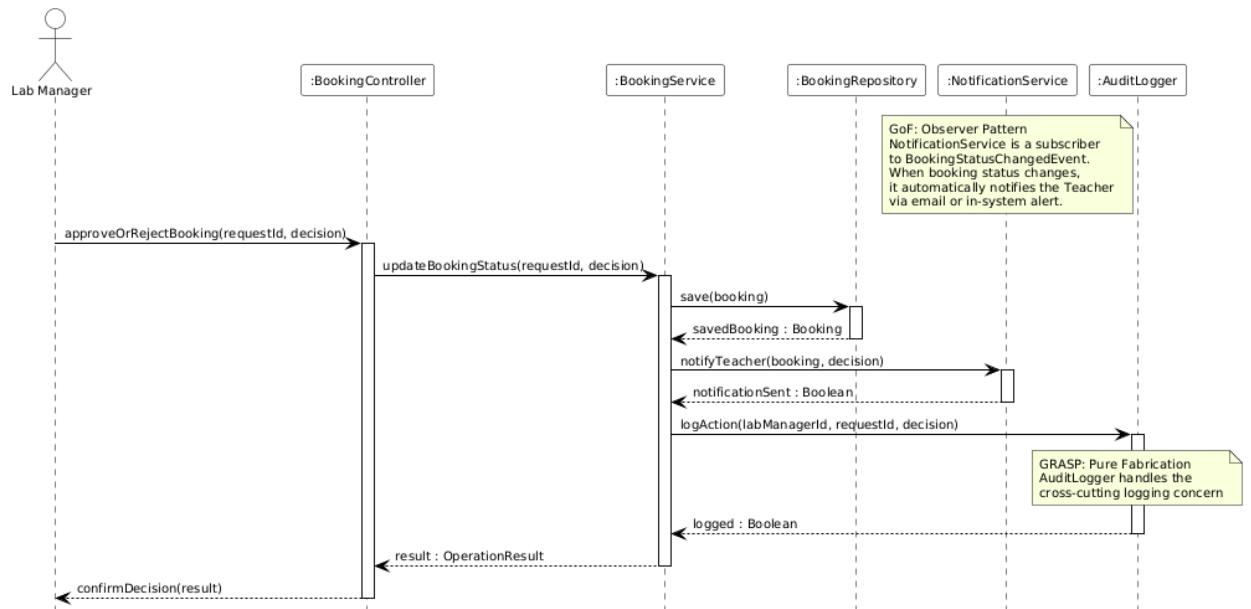
SD - UC01 Op2 : openBookingRequests()



SD - UC01 Op3 : checkAvailability(equipmentId, timeSlot)

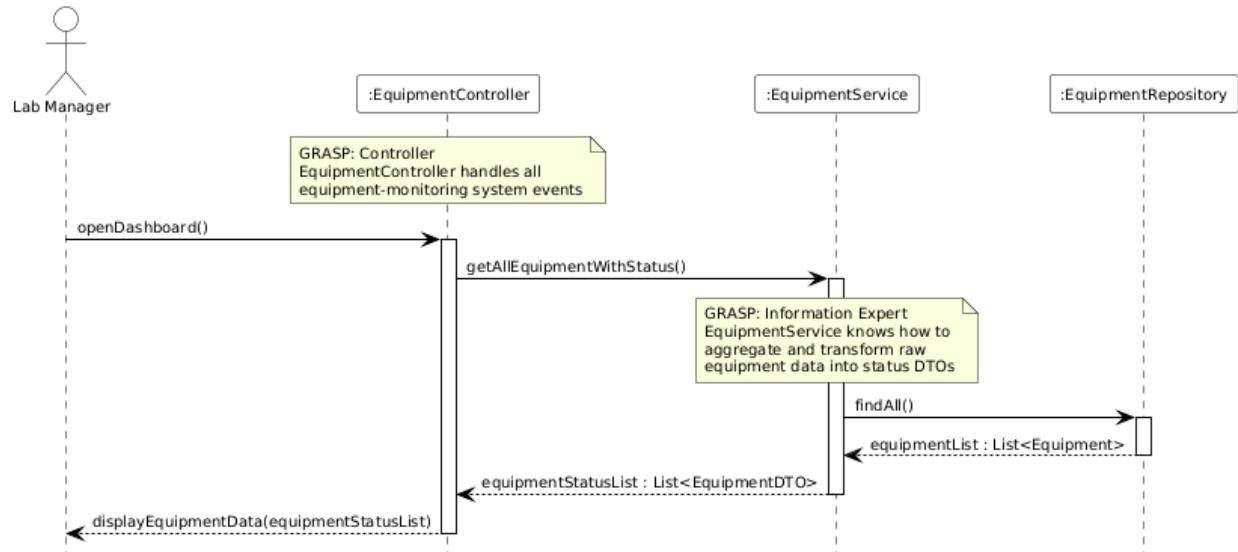


SD - UC01 Op4 : approveOrRejectBooking(requestId, decision)



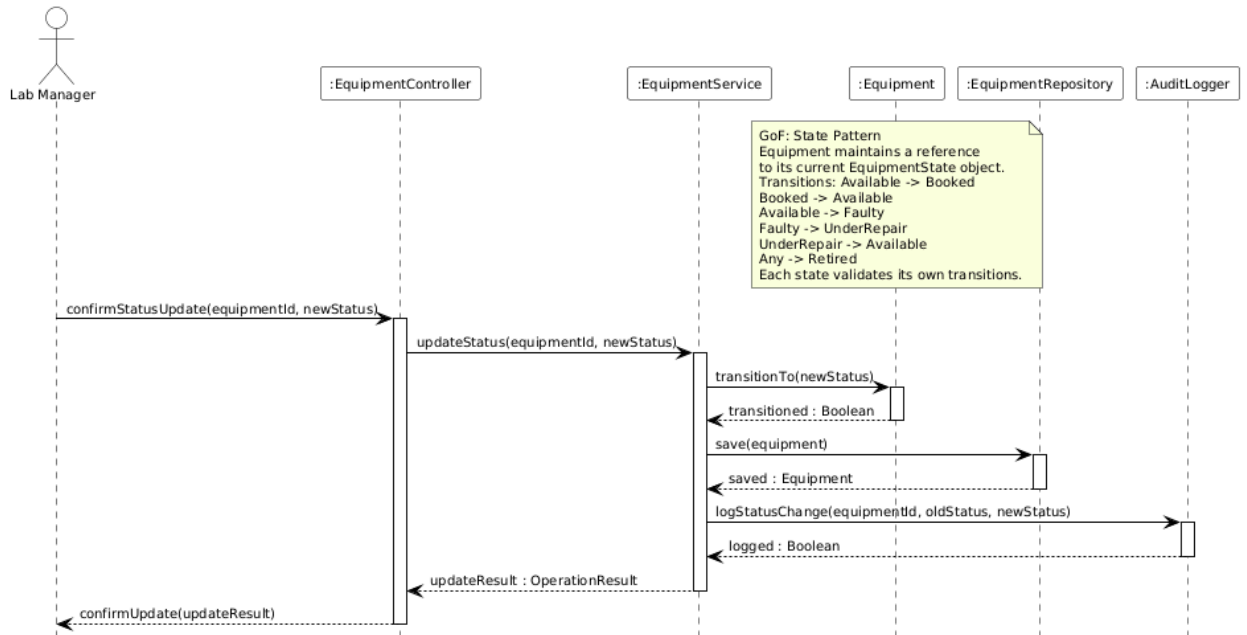
7-Monitor Equipments Availability

SD - UC02 Op1 : openDashboard()



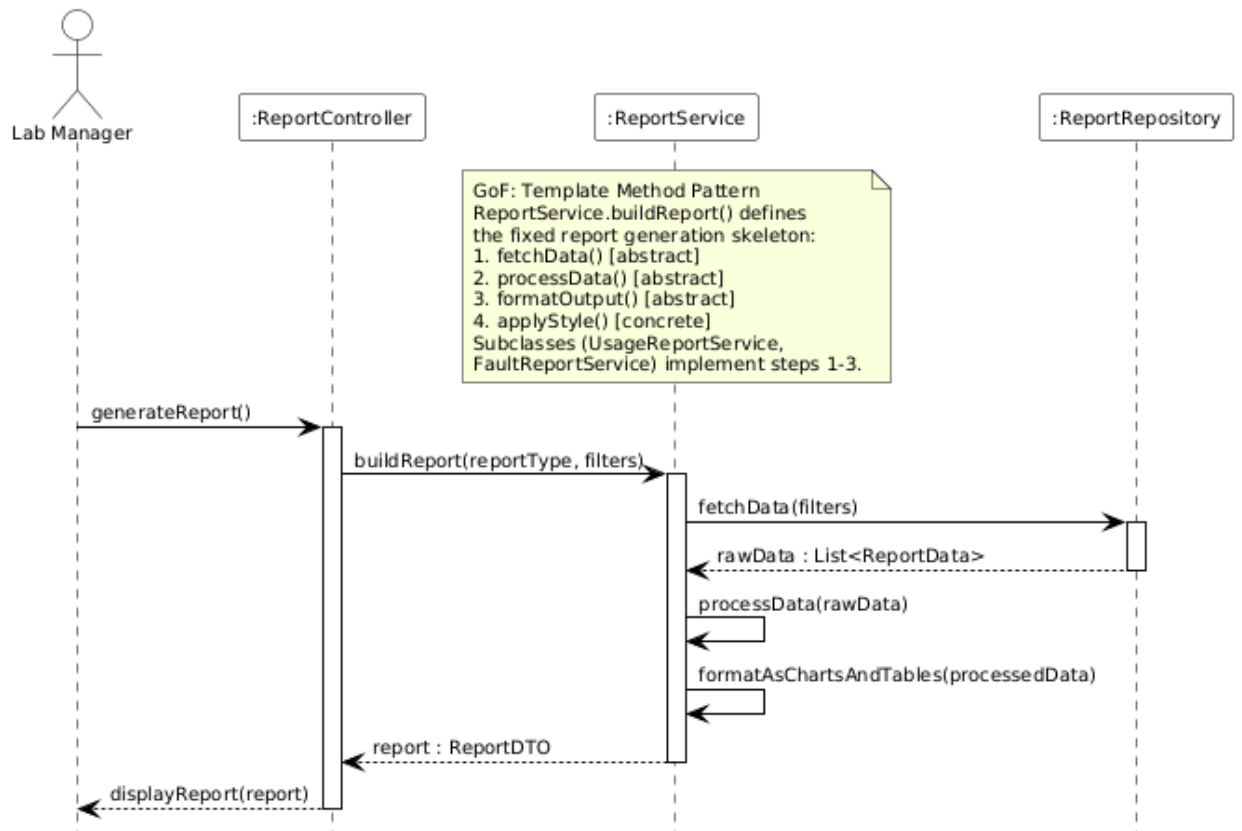
8-Update Equipment Status

SD - UC03 Op1 : confirmStatusUpdate(equipmentId, newStatus)



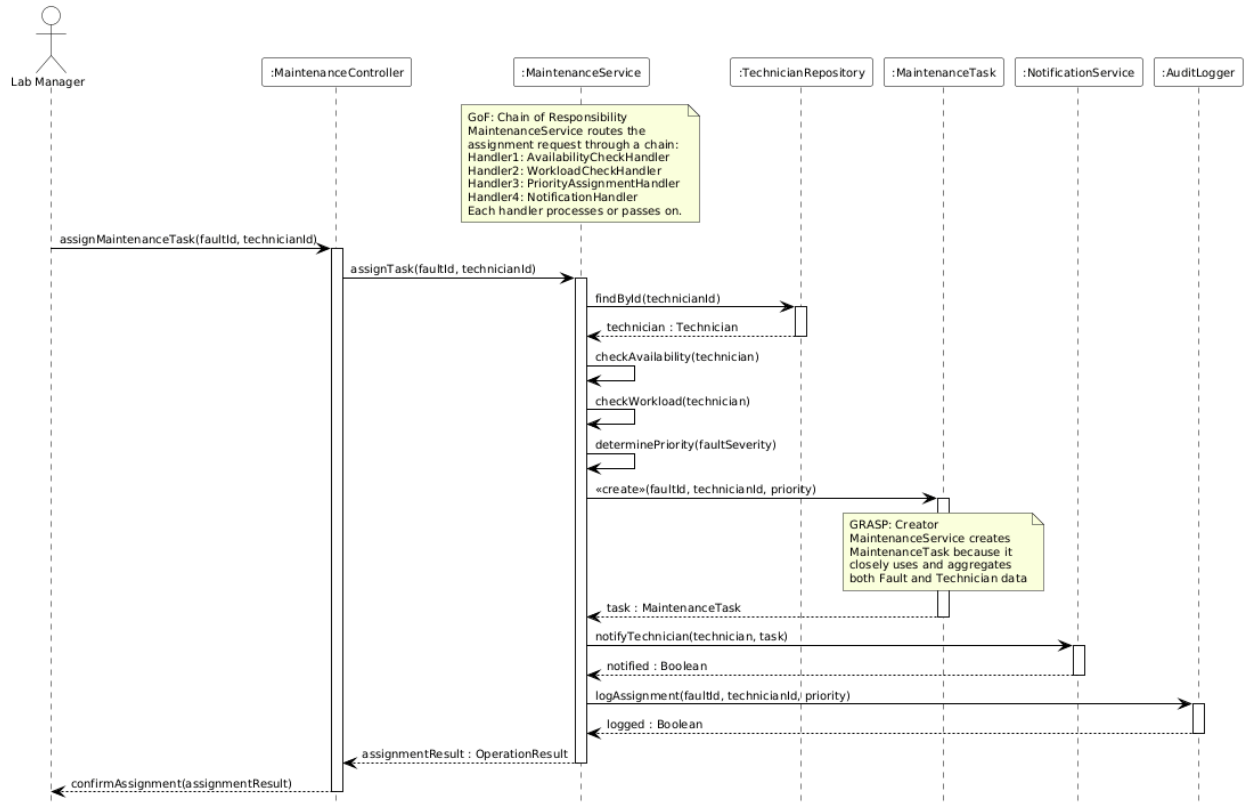
9-View Reports and Analytics

SD - UC04 Op1 : generateReport()

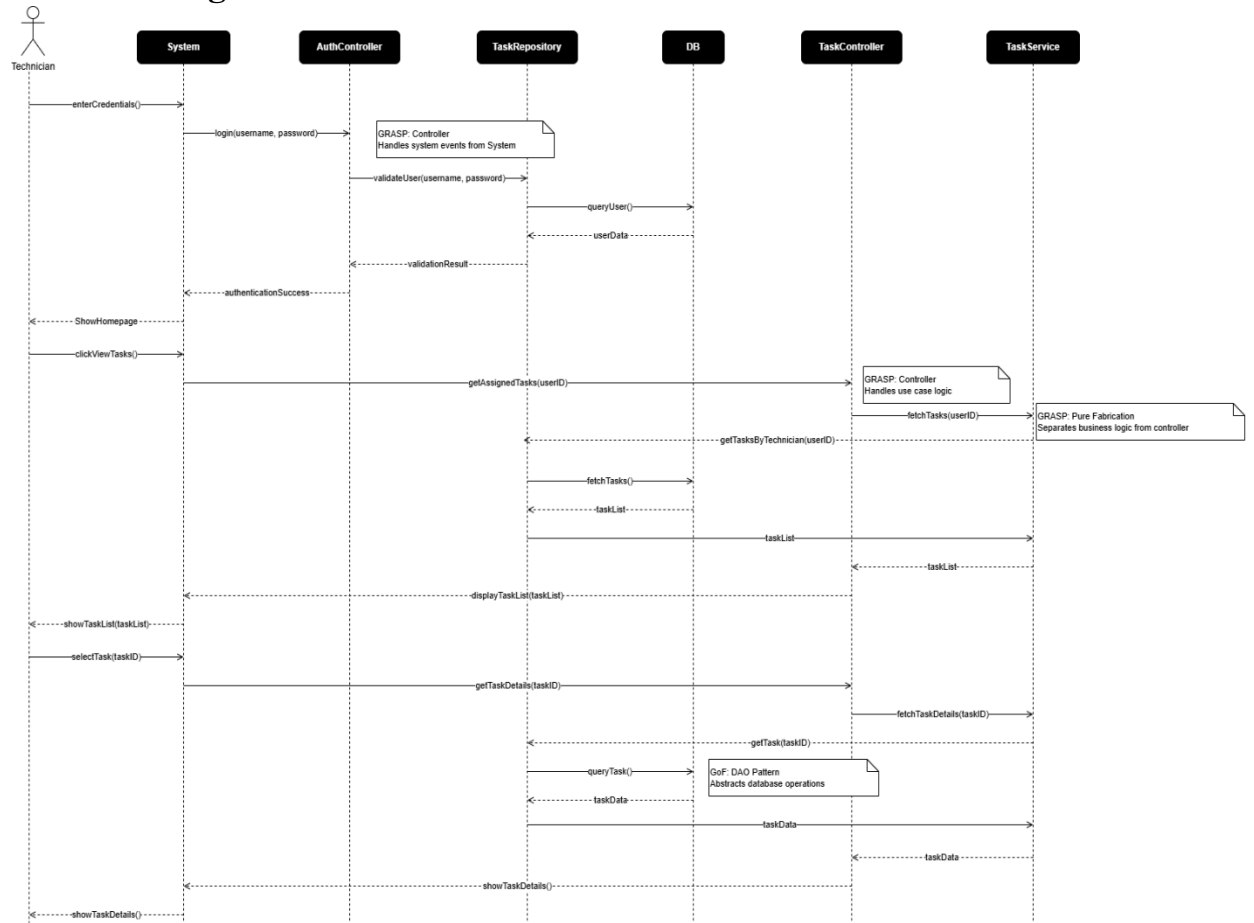


10-Assign Maintenance Tasks

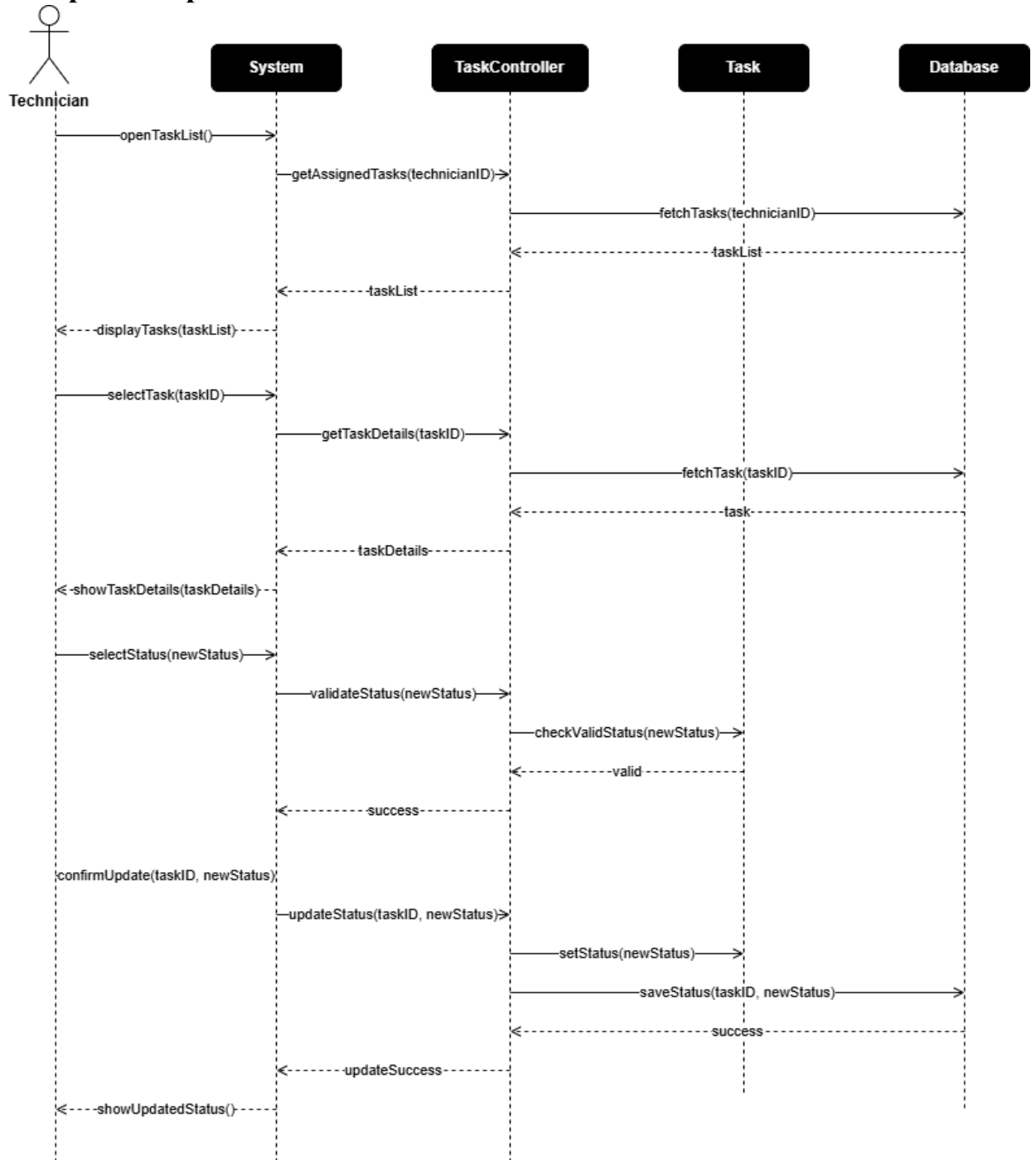
SD - UC05 Op1 : assignMaintenanceTask(faultId, technicianId)



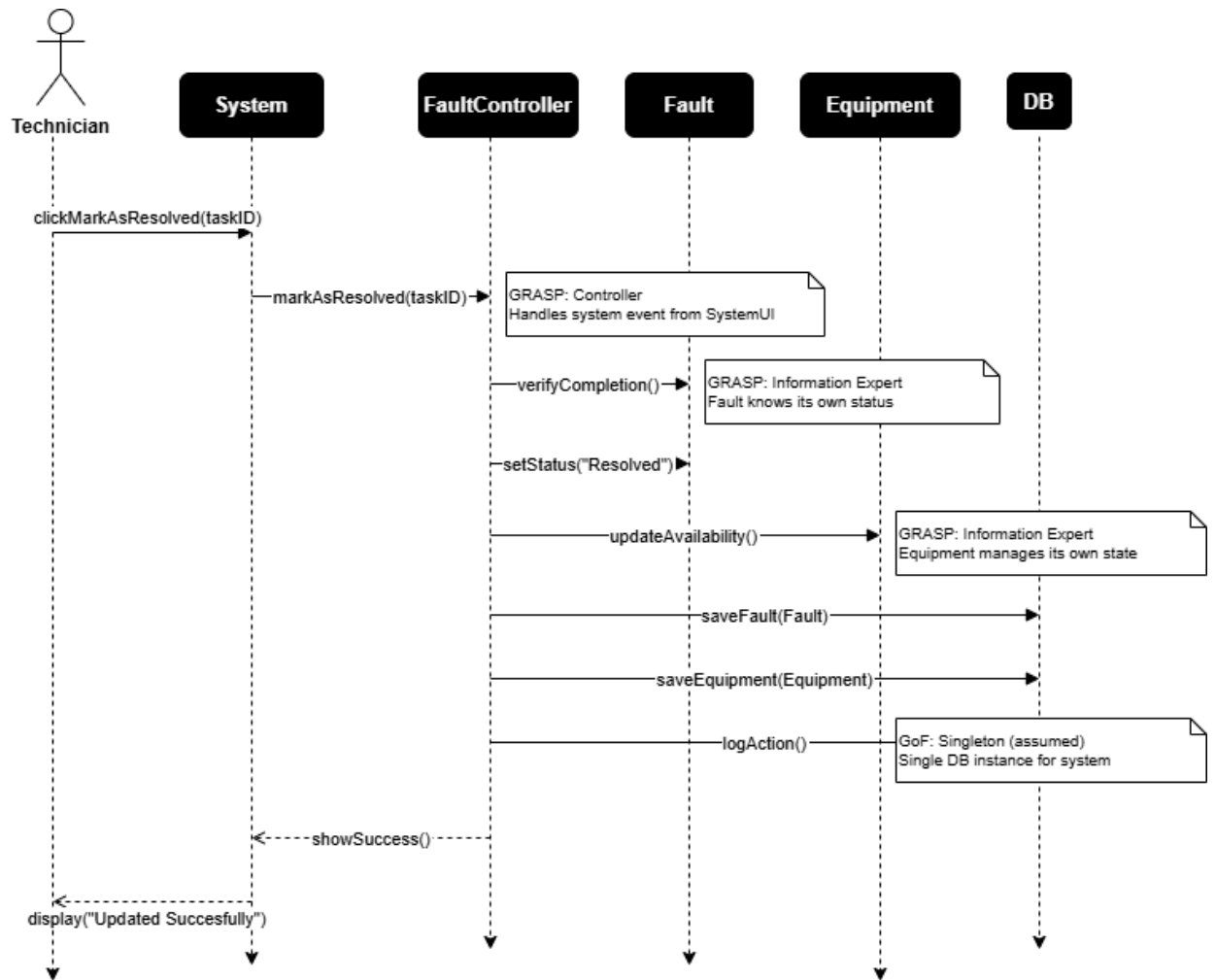
11-Veiw Assign Maintenance Task



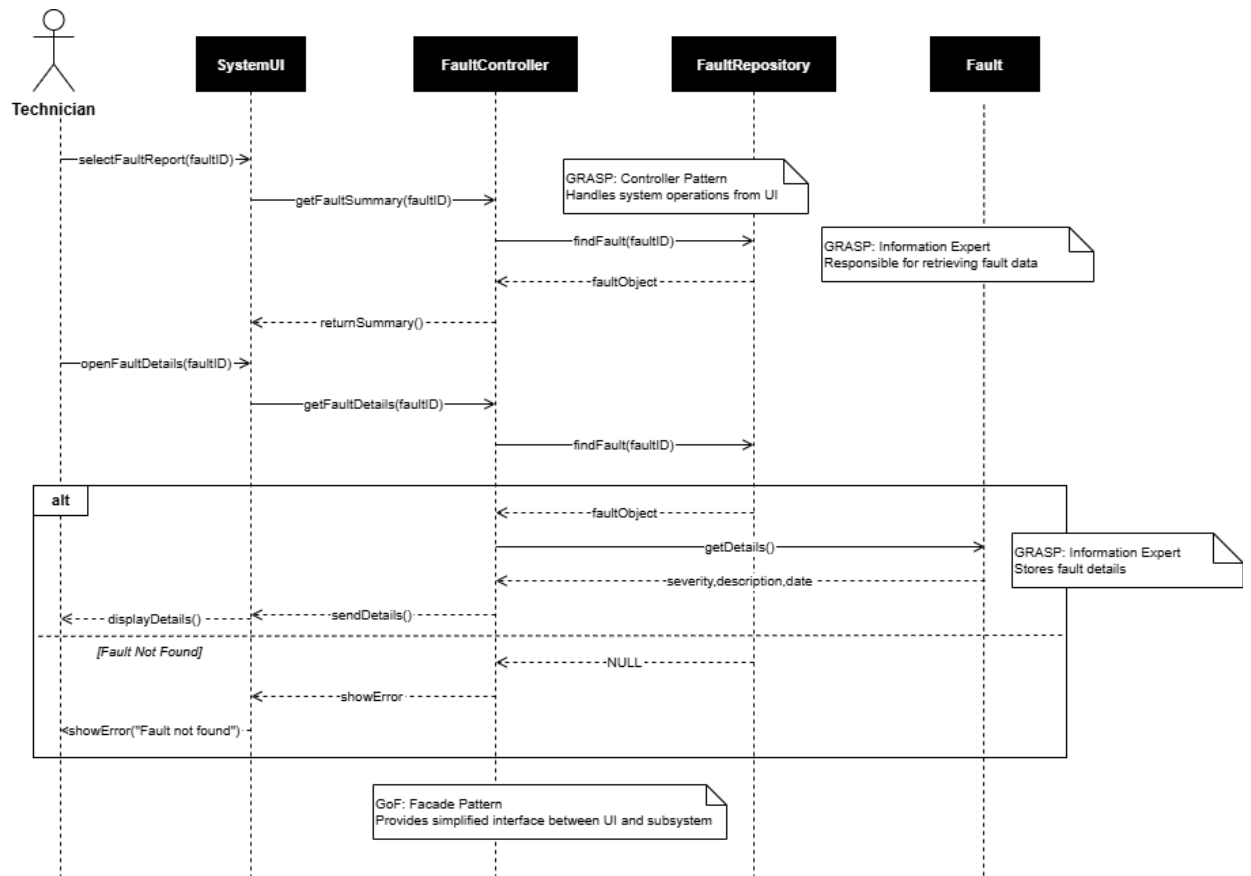
12-Update Repair Status



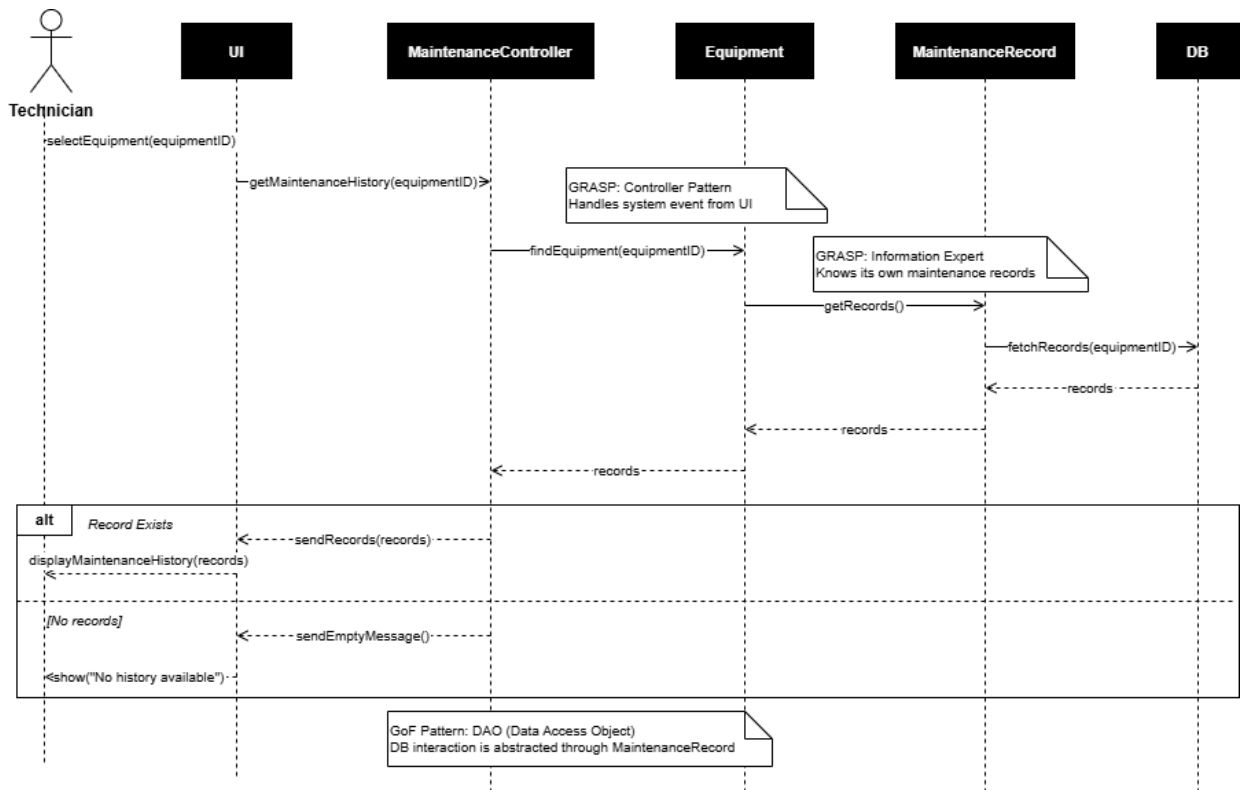
13-Mark Fault as Resolved



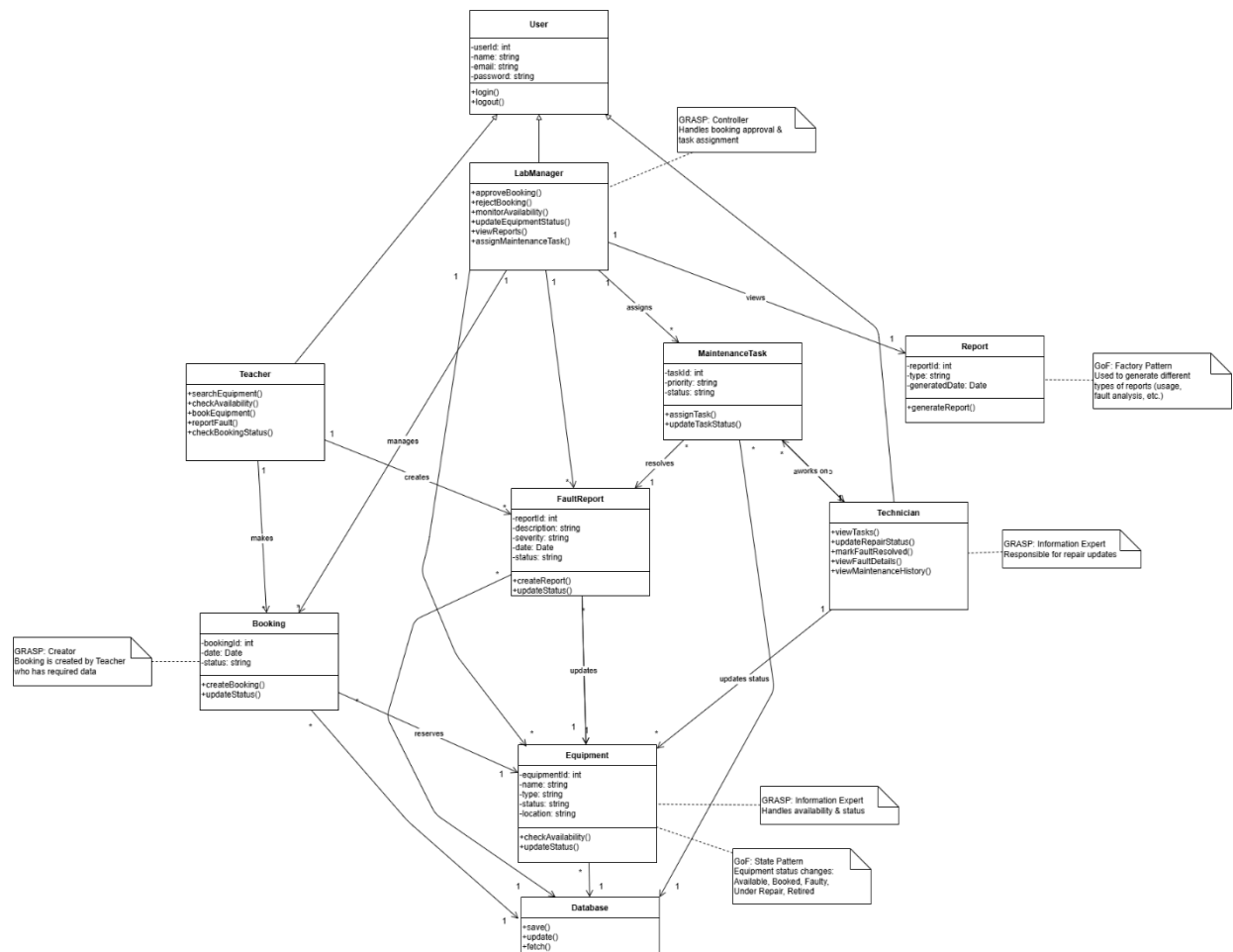
14-View Fault Details



15-View Maintenance History



7-Class Diagram



8. High-Level Architecture

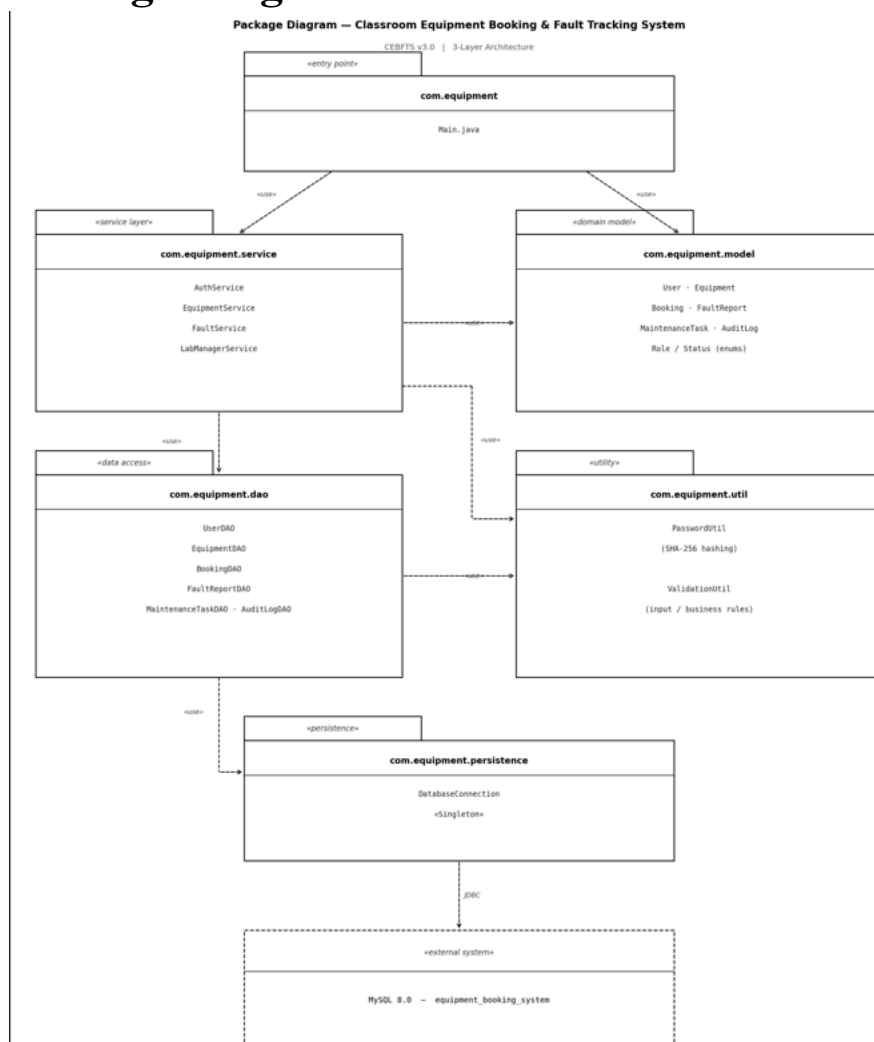
The system follows a 3-Layer Architecture (also known as the Three-Tier or Layered Architecture style).

Architectural Style Justification:

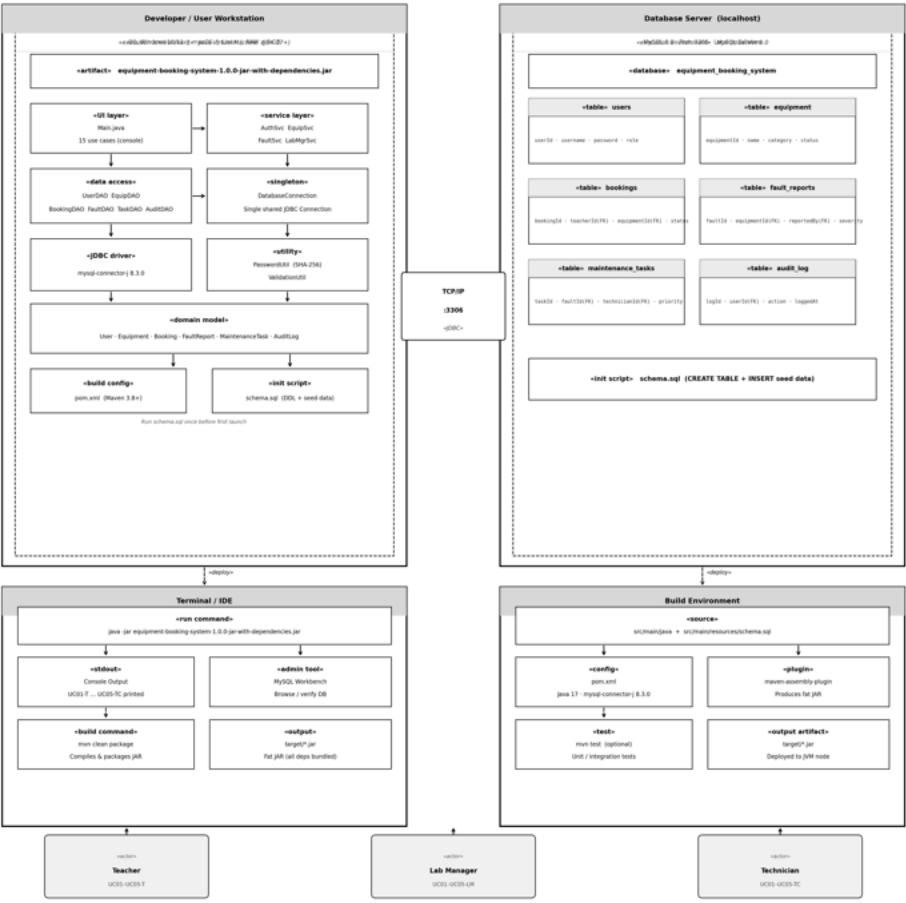
We selected the Layered (3-Tier) architectural style because it enforces separation of concerns, each layer only communicates with the layer directly below it, and changes in one layer (e.g., switching to a JavaFX UI) do not require changes to business logic or database code. This aligns well with academic software projects that emphasize OOP principles, testability, and maintainability.

Layer	Components	Responsibility
UI Layer	Main.java, style.css, dashboard_manager.fxml, Dashboard_teacher.fxml, dashboard_technician.fxml, login.fxml	Console-based user interaction. Invokes service methods. Displays output.
Business Logic Layer	AuthService.java, EquipmentService.java, FaultService.java, LabManagerService.java, PasswordUtil.java, ValidationUtil.java	Contains all application logic, validation, conflict checking, and workflow orchestration. Applies GRASP and GOF patterns.
Persistence Layer	UserDAO.java, EquipmentDAO.java, BookingDAO.java, FaultReportDAO.java, MaintenanceTaskDAO.java, AuditLogDAO.java, DatabaseConnection.java	All SQL queries and database interaction. Implements Singleton for connection management.

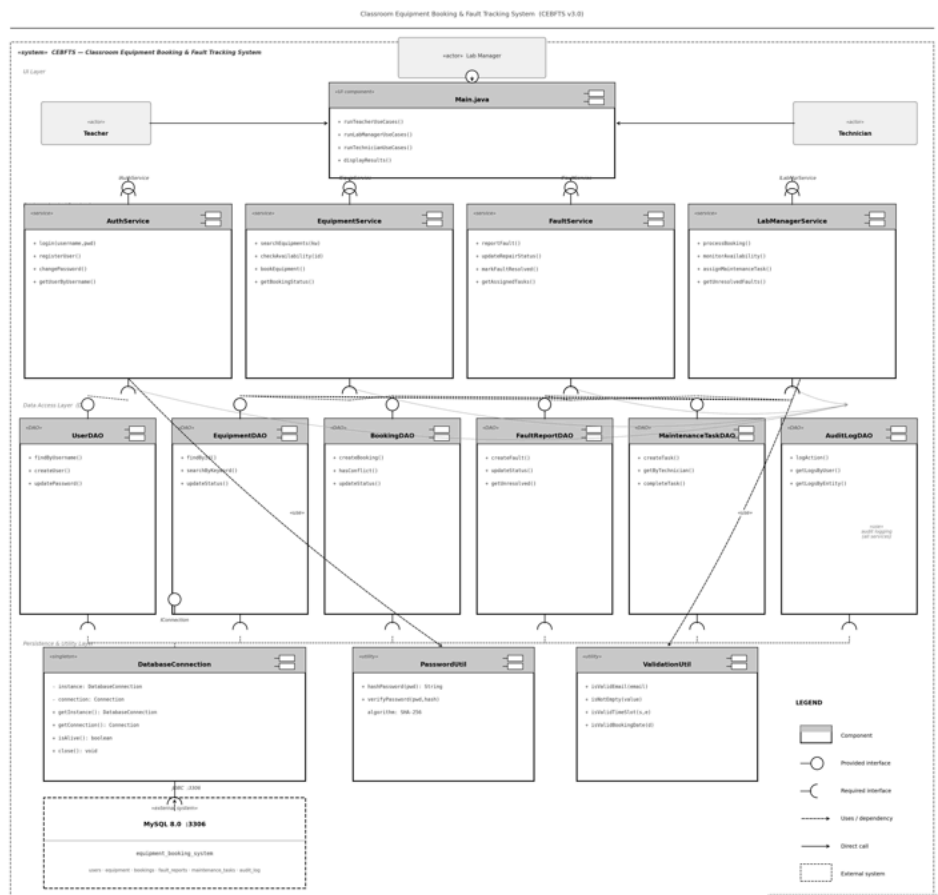
8.1 Package Diagram



8.2-Deployment Diagram



8.3-Component Diagram



9- Design Pattern

Pattern	Type	Where Applied	Justification
Singleton	GOF Creational	DatabaseConnection.java a	Ensures only one database connection is created and shared across all DAO classes, preventing resource exhaustion.

Facade	GOF Structural	AuthService, EquipmentService, FaultService, LabManagerService	Each service class provides a simplified interface to a complex subsystem of multiple DAOs, hiding multi-step coordination from the UI layer.
Controller (GRASP)	GRASP	All Service classes	Each service acts as the first point of contact for system operations related to its domain (auth, equipment, faults, management).
Creator (GRASP)	GRASP	Service classes	Service classes create their own DAO dependencies (constructor injection), fulfilling the Creator pattern since they use these objects extensively.
Information Expert (GRASP)	GRASP	DAO classes	Each DAO class has all the information necessary to handle persistence for its own entity type — it is the expert for its table.
Low Coupling / High Cohesion (GRASP)	GRASP	Entire architecture	Services are split by role/domain. No cross-service dependencies. DAOs handle exactly one entity each. UI layer has no SQL knowledge.

10. Work Division Table

Member	Role	Use Cases Implemented	Classes Implemented
Musfirah Hamid	Teacher use cases	UC01-T (Search), UC02-T (Check Availability), UC03-T (Book Equipment), UC05-T (Check Booking Status)	EquipmentService, EquipmentDAO, BookingDAO, Equipment.java, Booking.java
Saliha Noor	Lab Manager use cases	UC04-T (Report Fault), UC01-LM (Approve/Reject), UC02-LM (Monitor), UC03-LM (Update Status)	LabManagerService, FaultService (partial), FaultReportDAO, FaultReport.java
Sajal Ishtiaq	Technician + Manager use cases	UC04-LM (Reports), UC05-LM (Assign Task), UC01-TC to UC05-TC	MaintenanceTaskDAO, AuditLogDAO, AuthService, DatabaseConnection, Main.java

11. Database Schema

The SQL schema file (schema.sql) creates the following tables in the equipment_booking_system database:

Table	Primary Key	Foreign Keys	Key Columns
users	user_id (INT AI)	—	username (UNIQUE), password, full_name, email (UNIQUE), role ENUM(TEACHER, LAB_MANAGER, TECHNICIAN)
equipment	equipment_id (INT AI)	—	name, category, description, status ENUM(AVAILABLE, RESERVED, FAULTY, UNDER_REPAIR, RETIRED), location
bookings	booking_id (INT AI)	teacher_id → users, equipment_id → equipment, manager_id → users	start_datetime, end_datetime, purpose, status ENUM(PENDING, APPROVED, REJECTED, CANCELLED, COMPLETED)
fault_reports	fault_id (INT AI)	equipment_id → equipment, reported_by → users	fault_description, severity ENUM(LOW, MEDIUM, HIGH), status ENUM(REPORTED, ASSIGNED, IN_PROGRESS, RESOLVED), resolved date
maintenance_tasks	task_id (INT AI)	fault_id → fault_reports, technician_id → users, assigned_by → users	priority ENUM(LOW, MEDIUM, HIGH), status ENUM(PENDING, IN_PROGRESS, COMPLETED), notes, completed at
audit_log	log_id (INT AI)	user_id → users	action, entity_type, entity_id, details, logged at

12. Setup and Execution Instructions

Prerequisites

- Java 17 or higher (download from <https://adoptium.net>)
- Apache Maven 3.8 or higher
- Microsoft SQL Server 2017 or higher

Step 1: Database Setup

1. Open SSMS(SQL Server Management Studio) connected to LALEYKA\SQLEXPRESS.

2. Run the schema file: `source /path/to/equipment_project/src/main/resources/schema.sql`
3. This creates the database, all tables, and inserts sample seed data.

Step 2: Configure Database Connection

Verify the following settings match your SQL Server installation.

`SERVER = "YOUR_PC_NAME\\SQLEXPRESS"`

`DATABASE = "equipment_booking_system"`

`USER = "your_sql_username"`

`PASSWORD = "your_sql_password"`.

Step 3: Build and Run

```
cd equipment_project
```

```
mvn clean package
```

```
mvn javafx:run
```

Sample Login Credentials

Username	Password	Role
teacher1	hashed pass1	TEACHER
teacher2	hashed pass2	TEACHER
manager1	hashed pass3	LAB MANAGER
tech1	hashed pass4	TECHNICIAN
tech2	hashed pass5	TECHNICIAN